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(a) On which population, consisting of which observational units, are the data visualized here based?

Solution: Population / observational units: all U.S. states (+ DC). [1P.] Alternatively: (resident) population / inhabitants / citizens of the USA (1850–2010). ([1P.])

(b) Indicate the type of data collection and the survey form from which these data might originate. [2]

Solution: Complete survey (all U.S. states / all U.S. citizens). [1P.] Panel study / longitudinal data [1P.]

(c) Consider first only the part of the graphic labeled “AK” (= “Alaska”) at the top right next to the title. Analyze this part of the graphic with the terms of the *grammar of graphics* – state for all graphic means used in this part of the graphic which aesthetic mappings were used for which visualized variables.

Solution: Geom: area / polygon [1P.] [0.5P.] Mappings: x-pos: time [year] [1P.] y-pos or vertical extension: share of total population of Alaska (cumulative!) [1P.] [0.5P.] color: origin [1P.]

(d) What type of color scale is used here? Justify your answer. [2]

Solution: Qualitative. [1P.] Three different hues (gray – light/dark blue – orange) without inherent ordering. [1P.]

(e) Now consider the entire graphic for all states. Analyze the entire graphic using the terms from the grammar of graphics. Which *additional* graphical means were used to visualize which *additional* variables?

Solution: Graphical means: faceting by state. Additional variables: geographic position (longitude/latitude) of the states: x/y position of the facet for the state.

(f) For the features and statistics shown in the graphic, indicate the scale level. Briefly justify your answers in bullet points.

Solution:

- year: interval scale (metric but no natural zero point).
- state: nominal scale.
- origin: nominal or ordinal (both defensible).
- proportion: ratio scale (metric with natural zero point). Absolute scale also possible.
- geographical position (lat/lon: x-y position): interval scale.

(g) Which graphical means are appropriate or inappropriate in a given case depends on the communication goal underlying a statistical graphic. Use the terminology of the grammar of graphics or `ggplot2` pseudocode to describe how you would construct a graphic that answers the following question as simply and precisely as possible:

“How did the proportion of people born outside the USA change between 2000 and 2020 in the different states?”

Solution: For example:

- Geometry: points
- Mappings:
 - x-position: difference between proportion in 2000 and 2020
 - y-position: state

Two points for correct format (geometry + mappings), up to three points for a meaningful proposal.

(h) Indicate for the following statements whether they are true, not true, or cannot be decided based on the information in the graphic. Briefly justify your answers.

- (1) “In the years 1850 and 2020, about the same number of people in California (“CA”, 4th row from the top on the far left) lived who were born outside the USA.”
- (2) “The proportion of people in Florida (“FL”, bottom right), who were not born in Florida, was about the same in 2020 as it was in 1850.”

Solution:

- (1) Not decidable from the graphic – it shows proportions, not absolute numbers.
- (2) Yes – the width of the dark blue band in 2020 and 1850 is about the same.

2024/2

We have $N + 1$ piles $s_0, s_1, \dots, s_N$ of roulette chips with different proportions of red and blue chips. Each pile contains exactly N chips, and in pile s_j there are exactly j red chips and $N - j$ blue chips. From the $N + 1$ piles, one pile is chosen at random, and from this pile two chips are drawn without replacement.

Let $S_j, j \in \{0, \dots, N\}$ be the event that pile s_j is chosen. Let $R_i, i = 1, 2$ be the event that the i -th drawn chip is red.

- (a) Give $P(R_1 | S_j)$. What does this quantity mean intuitively?

Solution:

$$P(R_1 | S_j) = \frac{j}{N}.$$

This is the probability that the first chip drawn is red, given that the j -th pile was chosen.

- (b) Show that $P(R_1) = \frac{1}{2}$.

Solution:

$$\begin{aligned} P(R_1) &= \sum_{j=0}^N P(R_1 | S_j) P(S_j) \\ &= \sum_{j=0}^N \frac{j}{N} P(S_j) = \frac{1}{N+1} \sum_{j=0}^N j \\ &= \frac{1}{N(N+1)} \cdot \frac{N(N+1)}{2} = \frac{1}{2}. \end{aligned}$$

- (c) Determine $P(R_1 \cap R_2 | S_j)$. Simplify the expression as far as possible.

Solution:

$$P(R_1 \cap R_2 | S_j) = \begin{cases} \frac{\binom{j}{2} \binom{N-j}{0}}{\binom{N}{2}} = \frac{j(j-1)}{N(N-1)}, & j \geq 2, \\ 0, & j < 2. \end{cases}$$

- (d) Compute the (marginal) probability that both the first and the second drawn chip are red.

Solution. By the law of total probability and parts (a)–(c),

$$P(R_1 \cap R_2) = \sum_{j=0}^N P(R_1 \cap R_2 | S_j) P(S_j) = \frac{1}{N+1} \sum_{j=2}^N \frac{j(j-1)}{N(N-1)}.$$

Hence

$$P(R_1 \cap R_2) = \frac{1}{N(N+1)(N-1)} \sum_{j=2}^N j(j-1) = \frac{1}{N(N+1)(N-1)} \left(\sum_{j=1}^N j^2 - \sum_{j=1}^N j \right).$$

Using the standard sums,

$$\sum_{j=1}^N j = \frac{N(N+1)}{2}, \quad \sum_{j=1}^N j^2 = \frac{N(N+1)(2N+1)}{6},$$

we obtain

$$\begin{aligned} P(R_1 \cap R_2) &= \frac{1}{N(N+1)(N-1)} \left(\frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)}{2} \right) \\ &= \frac{1}{N(N+1)(N-1)} \cdot \frac{N(N+1)(2N-2)}{6}. \end{aligned}$$

Therefore,

$$P(R_1 \cap R_2) = \frac{2(N-1)}{6(N-1)} = \frac{1}{3}.$$

2024/3

(a) Three people each roll a standard fair die once in sequence. We are first interested in *all three rolls*. Give the smallest possible sample space Ω_1 (“smallest” here means a sample space that contains only outcomes that can actually occur, and no outcomes that cannot occur).

Solution. Any one of the following equivalent descriptions:

$$\begin{aligned} \Omega_1 &= \{(i, j, k) : i, j, k \in \{1, \dots, 6\}\} = \{1, \dots, 6\} \times \{1, \dots, 6\} \times \{1, \dots, 6\} \\ &= \{1, \dots, 6\}^3 = \{(1, 1, 1), (1, 1, 2), \dots, (6, 6, 6)\}. \end{aligned}$$

(b) Give the largest possible σ -algebra $\mathcal{F}_1$ on Ω_1 . By “largest” we mean the σ -algebra that contains the most sets, but does not include any sets that are not subsets of Ω_1 .

Solution.

$$\mathcal{F}_1 = \mathcal{P}(\Omega_1),$$

the power set of Ω_1 .

(c) You are now only interested in the sum of the three dice rolls (the “sum of the eyes”). Give the corresponding smallest possible sample space Ω_2 .

Solution.

$$\Omega_2 = \{3, 4, \dots, 17, 18\}$$

or an alternative solution such as

$$\Omega_2 = \{m : m = n_1 + n_2 + n_3, n_1, n_2, n_3 \in \{1, \dots, 6\}\}.$$

(d) Now give the following three σ -algebras for Ω_2 :

- (1) $\mathcal{F}_2$: the largest possible σ -algebra for Ω_2 (compare with (b))
- (2) $\mathcal{F}'_2$: the smallest possible σ -algebra (the σ -algebra containing the fewest elements)
- (3) $\mathcal{F}''_2$: a σ -algebra in which exactly one element contains exactly three elements. One example is sufficient.

Solution:

- $\mathcal{F}_2 = \mathcal{P}(\Omega_2)$, the power set of Ω_2 .
- $\mathcal{F}'_2 = \{\emptyset, \Omega_2\}$
- $\mathcal{F}''_2 = \{\emptyset, A, \Omega_2 \setminus A, \Omega_2\}$ with $A \subset \Omega_2$, $\text{card}(A) = 3$, and $A, \overline{A} \subseteq \Omega_2$. For example:

$$\mathcal{F}''_2 = \{\emptyset, \{3, 4, 5\}, \Omega_2 \setminus \{3, 4, 5\}, \Omega_2\}.$$

or any other valid example.

(e) Let $(\Omega_1, \mathcal{F}_1, \mu_Z)$ be a measure space with Ω_1 from part (a), $\mathcal{F}_1$ from (b), and μ_Z the counting measure. We now want to construct an image measure μ_f with the measure space $(\Omega_2, \mathcal{F}_2)$, such that

$$\mu_f : \mathcal{F}_2 \rightarrow [0, \infty[, \quad \mu_f(B) := \mu_Z(f^{-1}(B)).$$

Give the corresponding function $f : \Omega_1 \rightarrow \Omega_2$. (You do not need to show the measurability of f here.)

Solution:

$$f((i, j, k)) = i + j + k$$

Alternatively also possible, e.g.

$$\mathbf{x} = (x_1, x_2, x_3) \mapsto f(\mathbf{x}) = \sum_{i=1}^3 x_i$$

(f) Whether a function f is measurable or not has been shown in the lecture with the help of a table. You should also do this here, but not for all $B \in \mathcal{F}_2$, only for the two sets $\{3\}$ and $\{4\}$.

In addition to considering measurability, you should add another column in which you enter the corresponding value of $\mu_f(B)$. So your table should look something like this:

B	$f^{-1}(B)$	$\mu_f(B)$
$\{3\}$	$\{(1, 1, 1)\} \in \mathcal{F}_1 \checkmark$	1
$\{4\}$	$\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\} \in \mathcal{F}_1 \checkmark$	3

Solution: - For $B = \{3\}$ we have $f^{-1}(B) = \{(1, 1, 1)\}$, which is in $\mathcal{F}_1$, and $\mu_f(B) = 1$. - For $B = \{4\}$ we have $f^{-1}(B) = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$, which is in $\mathcal{F}_1$, and $\mu_f(B) = 3$.

2024/4

Let

$$F(x) = \begin{cases} 0, & x < 1, \\ 1 - ax^{-2}, & x \geq 1. \end{cases}$$

(1) Show that $F(x)$ for $a = 1$ is the distribution function of a continuous random variable. Necessary and sufficient: monotone increasing, continuous.

$$\lim_{x \searrow 1} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1$$

(1) **Monotone increasing:**

$$\frac{d}{dx} (1 - ax^{-2}) = 2ax^{-3} > 0 \quad \forall x > 0.$$

(2) **Continuity:** For $x > 1$ the function is continuous since it is a quadratic function in x^{-1} . Also at $x = 1$ we have

$$F(1) = 1 - 1 = 0 \quad \text{for } a = 1,$$

which matches the left-hand side limit.

(3) **Limits:** Since $\frac{a}{x^2} \rightarrow 0$ as $x \rightarrow \infty$, it follows that

$$F(x) \rightarrow 1 \quad \text{as } x \rightarrow \infty.$$

Thus, $F(x)$ with $a = 1$ is a valid distribution function of a continuous random variable.

2024/5A

We consider an exponentially distributed random variable X , with parameter $\lambda \in \mathbb{R}^+$. The density of X is defined as follows:

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

State with justification whether the random variable is continuous or discrete.

Solution:

Continuous, since the density is continuous on its support.

2024/5B

Let $Y := a + b \cdot X$, with $a, b \in \mathbb{R}^+$. Justify that Y is also a random variable.

Solution:

$g \circ X$ is the composition of measurable functions, hence $\mathcal{F} - \mathcal{B}$ measurable (Theorem 4.18) and thus a random variable (4.19). Alternatively, by Theorem 4.20: g is a continuous function, hence measurable.

2024/5C

Task. Let X be exponentially distributed with rate $\lambda > 0$, i.e.

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{(0, \infty)}(x).$$

For $a, b > 0$ define $Y = g(X) := a + bX$. Using the (distribution) transformation theorem, derive the cumulative distribution function F_Y of Y . Briefly verify the assumptions of the theorem and simplify the expression as far as possible.

Solution. Since $a, b > 0$, the map $g(x) = a + bx$ is one-to-one and strictly increasing. Hence g is invertible with

$$g^{-1}(y) = \frac{y - a}{b}.$$

Therefore, by the distribution transformation theorem,

$$F_Y(y) = \Pr(Y \leq y) = \Pr(X \leq g^{-1}(y)) = F_X\left(\frac{y - a}{b}\right).$$

Using $F_X(x) = (1 - e^{-\lambda x}) \mathbf{1}_{(0, \infty)}(x)$, we obtain

$$F_Y(y) = \left(1 - \exp\left[-\lambda \frac{y - a}{b}\right]\right) \mathbf{1}_{(a, \infty)}(y).$$

Equivalently, in piecewise form,

$$F_Y(y) = \begin{cases} 0, & y \leq a, \\ 1 - \exp\left(-\frac{\lambda}{b}(y - a)\right), & y > a. \end{cases}$$

2024/5D

Task. Give the density of

$$Z = \lambda \cdot X.$$

Which distribution does Z have?

Solution. Since $X \sim \text{Exp}(\lambda)$, applying the transformation $Z = \lambda X$ yields

$$F_Z(z) = P(Z \leq z) = P(\lambda X \leq z) = P\left(X \leq \frac{z}{\lambda}\right) = F_X\left(\frac{z}{\lambda}\right).$$

Thus

$$F_Z(z) = \left(1 - e^{-\lambda \cdot (z/\lambda)}\right) \mathbf{1}_{(0, \infty)}(z) = (1 - e^{-z}) \mathbf{1}_{(0, \infty)}(z).$$

Differentiating, the density is

$$f_Z(z) = e^{-z} \mathbf{1}_{(0, \infty)}(z).$$

Hence

$$Z \sim \text{Exp}(1).$$

2024/5E

Let $U \sim \text{Exp}(\nu)$ be another exponentially distributed random variable with parameter $\nu \in \mathbb{R}^+$. Assume U is stochastically independent of X . Furthermore, define $W = \nu U$ and the random variable D as follows:

$$D := W - Z := \nu U - \lambda X.$$

Show by means of convolution that the density of the random variable D is given by

$$f_D(d) = \frac{1}{2} \exp(-d) \mathbf{1}_{\mathbb{R}^+}(d).$$

Solution:

Convolution (Def. 15.24): the convolution $f * g$ of two functions is defined as

$$(f * g)(x) := \int_{\mathbb{R}_n} f(\tau) g(x - \tau) d\tau.$$

So, $D = W - Z \Rightarrow D + Z = W$.

$$\int_0^\infty f_Z(z) f_W(d + z) dz = \int_0^\infty \exp(-z) \exp(-(d + z)) dz$$

$$(1) \quad = \exp(-d) \int_0^\infty \exp(-2z) dz$$

$$(2) \quad = \exp(-d) \left[\left(-\frac{1}{2}\right) \exp(-2z) \right]_0^\infty$$

$$(3) \quad = \lim_{z \rightarrow \infty} \left(\left(-\frac{1}{2}\right) \exp(-d) \exp(-2z) \right) - \left(\left(-\frac{1}{2}\right) \exp(-d) \exp(-2 \cdot 0) \right)$$

$$(4) \quad = 0 + \left(\frac{1}{2} \exp(-d)\right)$$

$$(5) \quad = \frac{1}{2} \exp(-d).$$

Thus,

$$f_D(d) = \frac{1}{2} \exp(-d) \mathbf{1}_{\mathbb{R}^+}(d).$$

Remark: An alternative solution is possible using the density of $-\lambda U$, then performing the convolution of W and $(-Z)$.

2024/6A

At the train station of Hintertupfing, on average 3 trains stop per hour. The number of trains stopping in a given time interval is assumed to follow a Poisson distribution.

- (i) Under this distributional assumption, what is the probability that in a given 20-minute interval at least one train stops at Hintertupfing?
- (ii) A person arrives at the station exactly at the moment when a train has just left. Under the above distributional assumption, what is the probability that this person has to wait less than 60 minutes for the next train?

(i) Probability that at least one train arrives within 20 minutes.

The average number of trains per hour is 3. For a 20-minute interval (i.e., $\frac{1}{3}$ hour),

$$\lambda = 3 \cdot \frac{1}{3} = 1.$$

The number of trains in this interval is Poisson distributed:

$$X \sim \text{Poisson}(\lambda = 1).$$

Hence

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - e^{-\lambda} \frac{\lambda^0}{0!} = 1 - e^{-1} \approx 0.632.$$

(ii) Probability that the waiting time is less than 60 minutes.

Interarrival times of a Poisson process are exponentially distributed. With the rate measured per 20 minutes, we have

$$T \sim \text{Exp}(\lambda = 1).$$

Sixty minutes correspond to 3 intervals of 20 minutes, so

$$\mathbb{P}(T < 60 \text{ min}) = \mathbb{P}(T < 3) = F_T(3) = 1 - e^{-\lambda \cdot 3} = 1 - e^{-3} \approx 0.95.$$

Final answers:

$$\mathbb{P}(\text{at least one train in 20 minutes}) \approx 0.632, \quad \mathbb{P}(\text{waiting time} < 60 \text{ minutes}) \approx 0.95.$$

2024/6B

For the continuous random variable X with unimodal density $f_X(x)$, it is given that

$$\mathbb{E}(X) = 55, \quad F_X(35) = 0.5, \quad \text{Var}(X) = 9^2.$$

- (1) You draw a large random sample with independent values of X . Which is more likely: that the sample contains more outliers with particularly large values, or that it contains more outliers with particularly small values? Why?
- (2) Compute an upper bound for the probability that values of X occur which are smaller than 40 or larger than 70.

Solution:

- (1) The median is 35, the expectation is 55; hence the distribution is right-skewed (positively skewed). Therefore, more particularly large values are likely.
- (2) By Chebyshev's inequality:

$$P(|X - \mathbb{E}(X)| \geq 15) \leq \frac{\text{Var}(X)}{15^2} = \frac{9^2}{15^2} = \frac{9}{25} = 0.36.$$

2024/6C

A diagnostic test for a disease has a sensitivity of 95% and a specificity of 99%. The odds for having the disease in the population are 1 : 999. Compute the probability that a randomly selected person, who has been diagnosed with the disease, is actually sick.

Solution:

Let K : person is sick, and Pos : person tests positive.

$$\text{Odds } \gamma(K) = 1 : 999 \Rightarrow P(K) = 0.001.$$

By Bayes' theorem:

$$P(K | Pos) = \frac{P(Pos | K) \cdot P(K)}{P(Pos | K) \cdot P(K) + P(Pos | \bar{K}) \cdot P(\bar{K})}.$$

Substituting values:

$$= \frac{0.95 \cdot 0.001}{0.95 \cdot 0.001 + 0.01 \cdot 0.999} \approx \frac{0.00095}{0.01094} \approx 0.0868.$$

Thus, the probability is about 8.68%.

(Naturally, the denominator $P(Pos) = P(Pos | K) \cdot P(K) + P(Pos | \bar{K}) \cdot P(\bar{K}) \approx 0.01094$ can also be computed separately.)

2024/6D

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with expectation $\mu = 0 \in \mathbb{R}$ and variance σ^2 , $0 < \sigma^2 < \infty$. Consider the random variables $Y_1, Y_2, \dots$ defined by the transformation

$$Y_i := (X_i)^2.$$

Show that $\bar{Y}_n$ converges almost surely to a finite expectation; indicate which theorem you use, and verify the necessary conditions! First compute the corresponding expectation.

Solution:

The value to which $\bar{Y}_n$ converges almost surely is:

$$\mathbb{E}(Y_i) = \mathbb{E}((X_i)^2) = \mathbb{E}\left((X_i - \underbrace{\mathbb{E}(X_i)}_{=0})^2\right) = \text{Var}(X_i) = \sigma^2.$$

Proof of almost sure convergence: apply the **Strong Law of Large Numbers** (Kolmogorov).

Check the assumptions:

- $X_1, X_2, \dots$ are iid $\Rightarrow Y_1, Y_2, \dots$ are iid (since they are defined by the same transformation).
- $\mathbb{E}(Y_i) = \sigma^2 < \infty$, so the expectation exists.

Therefore, by the Strong Law of Large Numbers,

$$\bar{Y}_n \xrightarrow{\text{a.s.}} \mathbb{E}(Y_i) = \sigma^2.$$

2024/6E

Given the following covariance matrix candidate

$$A = \begin{pmatrix} 1 & 2 \\ a & b \end{pmatrix},$$

which values (single value or interval) can a and b take so that A is a covariance matrix?

Solution. A covariance matrix must be *symmetric* and *positive semidefinite (PSD)*.

- **Symmetry:** We must have $a = 2$.
- **Positive semidefiniteness:** For a 2×2 symmetric matrix, PSD is equivalent to the leading principal minors being nonnegative:

$$1 \geq 0 \quad \text{and} \quad \det(A) = \begin{vmatrix} 1 & 2 \\ 2 & b \end{vmatrix} = b - 4 \geq 0.$$

Hence $b \geq 4$.

Therefore, A is a covariance matrix if and only if

$$a = 2 \quad \text{and} \quad b \geq 4.$$

KLASAUR-1/3

Let the cumulative distribution function $F(x)$ be defined as:

$$F(x) = \begin{cases} 0 & x < 0 \\ \ln(ax + b) & 0 \leq x \leq \exp(1) \\ 1 & x > \exp(1) \end{cases}$$

- For which values of a and b is $F(x)$ the cumulative distribution function of a continuous random variable?
- What is the corresponding probability density function $f(x)$?
- What is the quantile function? Calculate the 95%-quantile.
- Is a distribution uniquely determined by $F(x)$? Justify your answer.
 - Is a distribution uniquely determined by $f(x)$? Justify your answer.

Solution to Problem 3

Let the cumulative distribution function $F(x)$ be defined as:

$$F(x) = \begin{cases} 0 & x < 0 \\ \ln(ax + b) & 0 \leq x \leq \exp(1) \\ 1 & x > \exp(1) \end{cases}$$

- For which values of a and b is $F(x)$ the cumulative distribution function of a continuous random variable?**

To be a valid cumulative distribution function (CDF), $F(x)$ must satisfy the following properties:

- $F(x)$ must be non-decreasing.
- $F(-\infty) = 0$ and $F(\infty) = 1$.
- $F(x)$ must be right-continuous.

The conditions for $F(x)$ to be a valid CDF are:

- For $F(x) = \ln(ax + b)$, the function must be non-decreasing for $0 \leq x \leq \exp(1)$, which requires that $a > 0$. - Additionally, $F(0) = \ln(b) = 0$, so $b = 1$.

Therefore, $a > 0$ and $b = 1$ are the values for which $F(x)$ is the CDF of a continuous random variable.

- What is the corresponding probability density function $f(x)$?**

The probability density function (PDF) $f(x)$ is the derivative of the CDF $F(x)$:

$$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx} \ln(ax + b) = \frac{a}{ax + b}$$

For $0 \leq x \leq \exp(1)$ and with $b = 1$, the PDF becomes:

$$f(x) = \frac{a}{ax + 1}$$

(c) **What is the quantile function? Calculate the 95%-quantile.**

The quantile function is the inverse of the CDF. For $0 \leq x \leq \exp(1)$, the CDF is $F(x) = \ln(ax + 1)$. To find the quantile function $Q(p)$, we solve $F(Q(p)) = p$:

$$\ln(aQ(p) + 1) = p \implies aQ(p) + 1 = e^p \implies Q(p) = \frac{e^p - 1}{a}$$

The 95%-quantile corresponds to $p = 0.95$:

$$Q(0.95) = \frac{e^{0.95} - 1}{a}$$

Therefore, the 95%-quantile is $Q(0.95) = \frac{e^{0.95} - 1}{a}$.

(d) (i) **Is a distribution uniquely determined by $F(x)$? Justify your answer.**

Yes, a distribution is uniquely determined by its cumulative distribution function $F(x)$. The CDF provides the complete information about the distribution, including the probabilities of all possible events. This is because the CDF fully defines the probability mass or density at every point.

(ii) **Is a distribution uniquely determined by $f(x)$? Justify your answer.**

No, a distribution is not uniquely determined by the probability density function $f(x)$ alone. The PDF $f(x)$ only provides information about the distribution where the random variable is continuous. It does not capture any discrete components of the distribution, nor does it provide information about the cumulative probabilities beyond the density function. To fully describe a distribution, both the CDF and the PDF may be needed.

KLASAUR-1/4

The company "Loysent" intends to implement a system for automatically detecting contaminated products as part of quality control in food production. The system is expected to check 5 million units per month during regular operation, and, on average, 10 out of every 1 million units are contaminated. In a pilot study of the system with a deliberately selected sample of products, the system raised an alarm for 13 out of 15 genuinely contaminated units and for 22 out of 1100 non-contaminated units.

(a) Based on the results of the pilot study, calculate the expected monthly frequencies of false alarms, correct alarms, missed contaminations, and correctly identified clean units, assuming the system is deployed in production.

Let:

- V : "Unit is contaminated"
- A : "System raises an alarm"

The system's performance in the pilot study gave us the following: - True positive rate (TPR/Sensitivity):

$$P(A | V) = \frac{13}{15} = 0.867$$

- False positive rate (FPR):

$$P(A | \bar{V}) = \frac{22}{1100} = 0.02$$

In regular operation, 5 million units are checked per month. Among these, 10 out of every 1 million are expected to be contaminated. Therefore, for 5 million units, we expect:

$$\frac{10}{1 \text{ million}} \times 5 \text{ million} = 50$$

contaminated units.

- **Correct alarms (true positives):**

The expected number of correct alarms is given by:

$$0.867 \times 50 = 43.33$$

- **Missed contaminations (false negatives):**

The expected number of missed contaminations is given by:

$$\frac{2}{15} \times 50 = 6.67$$

- **False alarms (false positives):**

Among the remaining 5 million – 50 = 4,999,950 non-contaminated units, the expected number of false alarms is:

$$0.02 \times 4,999,950 = 99,999$$

- **Correctly identified clean units (true negatives):**

The expected number of correctly identified clean units is:

$$(1 - 0.02) \times (5 \text{ million} - 50) = 0.98 \times 4,999,950 = 4,899,951$$

In summary:

- Correct alarms: 43.33
- Missed contaminations: 6.67
- False alarms: 99,999
- Correctly identified clean units: 4,899,951

Do you think the use of the system is statistically reasonable under the given conditions? Justify your answer quantitatively using appropriate metrics.

Solution:

We calculate the positive predictive value (PPV), which is the probability that a unit is contaminated given that the system raised an alarm. This is calculated as:

$$\begin{aligned} P(V | A) &= \frac{P(V \cap A)}{P(A)} = \frac{P(V \cap A)}{P(A \cap V) + P(A \cap \bar{V})} \\ &= \frac{P(A | V) \cdot P(V)}{P(A | V) \cdot P(V) + P(A | \bar{V}) \cdot P(\bar{V})} \end{aligned}$$

Substitute the known values: - $P(A | V) = \frac{13}{15}$ - $P(V) = \frac{10}{10^6}$ - $P(A | \bar{V}) = \frac{22}{1100}$ - $P(\bar{V}) = 1 - P(V) = 1 - \frac{10}{10^6}$

$$P(V | A) = \frac{\frac{13}{15} \cdot \frac{10}{10^6}}{\frac{13}{15} \cdot \frac{10}{10^6} + \frac{22}{1100} \cdot \left(1 - \frac{10}{10^6}\right)}$$

Simplifying:

$$P(V | A) = \frac{\frac{13}{15} \cdot 10^{-5}}{\frac{13}{15} \cdot 10^{-5} + \frac{22}{1100}} = 4.3 \times 10^{-5}$$

Thus, the positive predictive value (PPV) is approximately 4.3×10^{-5} .

(Alternatively, using the contingency table from earlier, we can compute: $PPV = \frac{43}{100042}$).

The PPV is very close to 0, meaning that a positive test result provides very little information, and the overwhelming majority of alarms are false positives. The test is too imprecise to be used effectively for automatic monitoring of the entire production process.

KLASAUR-1/5

Given $\Omega = \{1, 2, 3, 4\}$. Let $A = \{1\}$, $B = \{1, 3\}$, $\mu(A) := \sum_{\omega \in A} \omega$, and $f : \Omega \rightarrow \mathbb{R}$ with $f(\omega) = 1/\omega$.

- Specify $\mathcal{F}_1 = \sigma(\{A, B\})$.
- Show that the union of two sigma-algebras does not necessarily form a sigma-algebra. Define a sigma-algebra $\mathcal{F}_2$ for Ω such that $\mathcal{F}_1 \cup \mathcal{F}_2$ is not a sigma-algebra.
- Show that μ is a measure on the measurable space $(\Omega, \mathcal{F}_1)$.
- Compute $\int_B f d\mu$.
- Is f the density of a distribution with respect to the dominating measure μ ? Justify your answer.

Solution Elementary sets are $A = \{1\}$, $B \cap \bar{A} = \{3\}$, $\bar{A} \cap \bar{B} = \{2, 4\}$. These three sets: $\{1\}$, $\{3\}$, $\{2, 4\}$ are disjoint and together cover Ω . They are called the elementary sets (or atoms) generated by A and B. Then $\sigma(\{A, B\}) = \mathcal{P}(\{\{1\}, \{3\}, \{2, 4\}\})$ and has 8 elements.

Part (d) On a finite discrete space the integral is a weighted sum:

$$\int_B f d\mu = \sum_{\omega \in B} f(\omega) \mu(\{\omega\}).$$

Since $\mu(\{\omega\}) = \omega$,

$$\int_B f d\mu = f(1) \cdot 1 + f(3) \cdot 3 = \frac{1}{1} \cdot 1 + \frac{1}{3} \cdot 3 = 1 + 1 = 2.$$

Part (e).

A function f is a probability density w.r.t. a dominating measure μ iff $f \geq 0$ μ -a.e. and

$$\int_{\Omega} f d\mu = 1.$$

Here $\Omega = \{1, 2, 3, 4\}$, $\mu(\{\omega\}) = \omega$, and $f(\omega) = 1/\omega$. Thus

$$\int_{\Omega} f d\mu = \sum_{\omega \in \Omega} f(\omega) \mu(\{\omega\}) = \sum_{\omega \in \Omega} \frac{1}{\omega} \cdot \omega = \sum_{\omega \in \Omega} 1 = 4 \neq 1.$$

Hence f is *not* the density of a probability distribution with respect to μ . (Indeed, f yields the counting measure: $\int_A f d\mu = |A|$ for all $A \subseteq \Omega$. A valid probability density would be $\tilde{f} = \frac{1}{4}f$, for which $\int_{\Omega} \tilde{f} d\mu = 1$.)

KLASAUR-1/6

Given

$$f(x) = \frac{3}{2}(x-1)^2 \cdot I_{[0,2]}(x)$$

the continuous density of the random variable X .

- Sketch the density. What is the mode of the distribution? Provide a reasoning for the expectation, median, and skewness of the distribution without calculation.
- Calculate the variance of the distribution.
- Calculate the density of $Y = X^2$.
- Estimate the expectation of $Z = \sin(X)$. (No explicit calculation!)

The Mode of a Random Variable

The **mode** of a random variable is the value (or values) at which the probability density function (for continuous variables) or the probability mass function (for discrete variables) attains its maximum.

Part (c) - Density of $Y = X^2$

We are given that X has the probability density function:

$$f_X(x) = \frac{3}{2}(x-1)^2 \cdot I_{[0,2]}(x)$$

Now, we want to find the density function of $Y = X^2$.

Step 1: Change of variables

We perform a change of variables. Let $Y = X^2$, so that $X = \sqrt{Y}$. Since $X \in [0, 2]$, this implies that $Y \in [0, 4]$.

Now, differentiate $Y = X^2$ with respect to x :

$$\frac{dy}{dx} = 2x$$

Thus, the derivative of the inverse function is:

$$\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

Step 2: Transformation of the density

The density function of Y , denoted $f_Y(y)$, can be found using the formula for change of variables in probability densities:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute $x = \sqrt{y}$ and $\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$:

$$f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}}$$

Substitute $f_X(x) = \frac{3}{2}(x-1)^2 \cdot I_{[0,2]}(x)$ into the equation:

$$f_Y(y) = \frac{3}{2}(\sqrt{y}-1)^2 \cdot I_{[0,4]}(y) \cdot \frac{1}{2\sqrt{y}}$$

Simplifying the expression:

$$f_Y(y) = \frac{3}{4} \cdot \frac{(\sqrt{y}-1)^2}{\sqrt{y}} \cdot I_{[0,4]}(y)$$

Thus, the density of $Y = X^2$ is:

$$f_Y(y) = \frac{3}{4} \cdot \frac{(\sqrt{y}-1)^2}{\sqrt{y}} \cdot I_{[0,4]}(y)$$

Part (d)

Since $\sin(x)$ is concave on $[0, 2]$, we can apply Jensen's inequality, which states that for a concave function φ :

$$\varphi(\mathbb{E}[X]) \geq \mathbb{E}[\varphi(X)]$$

In this case, the inequality becomes:

$$\sin(\mathbb{E}[X]) \geq \mathbb{E}[\sin(X)]$$

From part (a), we know that $\mathbb{E}[X] = 1$. Therefore, applying Jensen's inequality gives:

$$0.841 \approx \sin(1) \geq \mathbb{E}[\sin(X)]$$

KLASAUR-1/7

Prof. S. takes the bus to the university every day ($n = 225$) and is always on time at the bus stop. However, the bus is late every day. Let X_i denote the random delay time in minutes on day i . Assume that X_i , for $i = 1, \dots, n$, are independent and identically exponentially distributed with $\mathbb{E}(X_i) = 1$.

Consider the random variable $X = \sum_{i=1}^n X_i$, which is the total delay in one year.

- (a) Show that X is approximately normally distributed. What are the approximate expected value and approximate variance?

Solution

Since $\mathbb{E}(X_i) = \mu = 1 = \frac{1}{\lambda}$, we have $\lambda = 1$ and thus $\text{Var}(X_i) = \frac{1}{\lambda^2} = 1 = \sigma^2 < \infty$. Since the X_i are independent and identically distributed, by the Central Limit Theorem:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{a} \mathcal{N}(0, 1)$$

$$\Leftrightarrow \bar{X} - \mu \xrightarrow{a} \mathcal{N}\left(0, \frac{\sigma^2}{n}\right)$$

$$\Leftrightarrow \frac{1}{n} X \xrightarrow{a} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\Leftrightarrow X \xrightarrow{a} \mathcal{N}(n\mu, n\sigma^2)$$

With $\mu_X = n\mu = 225 \cdot 1 = 225$ and $\sigma_X^2 = n\sigma^2 = 225 \cdot 1^2 = 225$.

Determine approximately the probability that Prof. S. waits more than 4 hours for the bus over the course of the year. *Hint:* $\Phi(0.5) \approx 0.6915$, $\Phi(1) \approx 0.8413$, $\Phi(\sqrt{2}) \approx 0.9214$, $\Phi(2) \approx 0.9772$.

Solution:

We are interested in:

$$\begin{aligned} \mathbb{P}(X > 4 \cdot 60) &\equiv \mathbb{P}(X > 240) = 1 - \mathbb{P}(X \leq 240) \\ &\equiv 1 - \mathbb{P}\left(\frac{X - \mu_X}{\sigma_X} \leq \frac{240 - 225}{15}\right) \\ &\approx 1 - \mathbb{P}\left(\frac{X - \mu_X}{\sigma_X} \leq 1\right) \\ &\approx 1 - \Phi(1) \approx 1 - 0.84 = 0.16 \end{aligned}$$

KLASAUR-1/8

Let $\mathbf{X} = (X_1, X_2)$ be a random vector with joint density

$$f(x_1, x_2) = \begin{cases} \lambda \exp\left(-\sqrt{\lambda}(x_1 + x_2)\right) & \text{for } x_1 > 0 \text{ and } x_2 > 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that X_1 and X_2 are stochastically independent.
 (b) Derive the marginal densities of X_1 and X_2 . Are these well-known distributions?
 (c) Derive the density of $Y = X_1 + X_2$.

Part (a) - Show that X_1 and X_2 are stochastically independent

To show that X_1 and X_2 are stochastically independent, we need to demonstrate that the joint density $f(x_1, x_2)$ factorizes into the product of the marginal densities $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$. Given the joint density:

$$f(x_1, x_2) = \lambda \exp\left(-\sqrt{\lambda}(x_1 + x_2)\right) \quad \text{for } x_1 > 0 \text{ and } x_2 > 0,$$

we can rewrite this as:

$$f(x_1, x_2) = \left(\sqrt{\lambda} \exp\left(-\sqrt{\lambda}x_1\right)\right) \cdot \left(\sqrt{\lambda} \exp\left(-\sqrt{\lambda}x_2\right)\right).$$

The marginal density of X_1 is:

$$f_{X_1}(x_1) = \int_0^\infty \lambda \exp\left(-\sqrt{\lambda}(x_1 + x_2)\right) dx_2 = \sqrt{\lambda} \exp(-\sqrt{\lambda}x_1)$$

This shows how you "marginalize out" X_2 to focus on the marginal density of X_1 . Thus, the joint density factorizes as:

$$f(x_1, x_2) = f_{X_1}(x_1) \cdot f_{X_2}(x_2),$$

where

$$f_{X_1}(x_1) = \sqrt{\lambda} \exp\left(-\sqrt{\lambda}x_1\right), \quad f_{X_2}(x_2) = \sqrt{\lambda} \exp\left(-\sqrt{\lambda}x_2\right).$$

Since the joint density factorizes, we conclude that X_1 and X_2 are stochastically independent.

Part (b) - Derive the marginal densities of X_1 and X_2

From part (a), we already see that the marginal densities $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ are:

$$f_{X_1}(x_1) = \sqrt{\lambda} \exp\left(-\sqrt{\lambda}x_1\right), \quad x_1 > 0,$$

and

$$f_{X_2}(x_2) = \sqrt{\lambda} \exp\left(-\sqrt{\lambda}x_2\right), \quad x_2 > 0.$$

These are exponential distributions with rate parameter $\sqrt{\lambda}$. Therefore, both X_1 and X_2 follow independent exponential distributions:

$$X_1, X_2 \sim \text{Exp}(\sqrt{\lambda}).$$

Part (c) - Derive the density of $Y = X_1 + X_2$

We want to derive the density of $Y = X_1 + X_2$, where X_1 and X_2 are independent exponential random variables with rate $\sqrt{\lambda}$.

The sum of two independent exponential random variables with the same rate μ follows a Gamma distribution with shape parameter 2 and rate μ . In this case, the rate is $\sqrt{\lambda}$, so:

$$Y = X_1 + X_2 \sim \text{Gamma}(2, \sqrt{\lambda}).$$

The probability density function of a Gamma distribution with shape parameter $k = 2$ and rate $\theta = \sqrt{\lambda}$ is given by:

$$f_Y(y) = \frac{(\sqrt{\lambda})^2}{(2-1)!} y^{2-1} e^{-\sqrt{\lambda}y}, \quad y > 0.$$

Simplifying:

$$f_Y(y) = \lambda y e^{-\sqrt{\lambda}y}, \quad y > 0.$$

Thus, the density of $Y = X_1 + X_2$ is:

$$f_Y(y) = \lambda y e^{-\sqrt{\lambda}y}, \quad y > 0.$$

Method 2 (CDF then differentiate).

$$\begin{aligned} F_Y(y) &= \mathbb{P}(X_1 + X_2 \leq y) = \int_0^y \mathbb{P}(X_2 \leq y - x) f_{X_1}(x) dx = \int_0^y (1 - e^{-\sqrt{\lambda}(y-x)}) \sqrt{\lambda} e^{-\sqrt{\lambda}x} dx \\ &= 1 - e^{-\sqrt{\lambda}y} (1 + \sqrt{\lambda}y), \quad \Rightarrow \quad f_Y(y) = F'_Y(y) = \lambda y e^{-\sqrt{\lambda}y}, \quad y > 0. \end{aligned}$$

KLASAUR-1/9

Let $\mathbf{X} = (X_1, X_2, X_3)$ be multivariate normally distributed with mean $\mu = (4, 1, 3)$ and covariance matrix

$$\Sigma = \begin{pmatrix} 2 & 0.5 & -2 \\ A & 0.5 & 0 \\ -2 & 0 & B \end{pmatrix}$$

- Determine A and justify your answer. Also, provide a justification for the highest possible lower bound for B .
 - What is the distribution of $Y = \sum_{i=1}^3 X_i$? What are the expectation and variance of Y ? Justify your answer.
 - Calculate $\mathbb{E}[\exp(X_3) \cdot (X_2 - 1)]$ and justify your answer.
-

Solution:

- Determine A and justify your answer. Also, provide a justification for the highest possible lower bound for B .**

Since the covariance matrix Σ is symmetric, we know that $A = 0.5$.

To find the lower bound for B , we use the condition that the determinant of the covariance matrix must be non-negative. Calculating the determinant of Σ :

$$\det(\Sigma) = B - 2 - \frac{1}{4}B \geq 0$$

Simplifying this inequality gives:

$$B - 2 \geq \frac{1}{4}B \quad \Rightarrow \quad \frac{3}{4}B \geq 2 \quad \Rightarrow \quad B \geq \frac{8}{3}$$

Hence, the highest possible lower bound for B is $B \geq \frac{8}{3}$.

Another way to determine B is by considering the correlation coefficient ρ_{13} , which is given by:

$$\rho_{13} = \frac{\text{Cov}(X_1, X_3)}{\sqrt{\text{Var}(X_1)\text{Var}(X_3)}} = \frac{-2}{\sqrt{2B}}$$

Since $|\rho_{13}| \leq 1$, we get:

$$\frac{-2}{\sqrt{2B}} \geq -1 \quad \Rightarrow \quad B \geq 2$$

Therefore, the lower bound for B is $B \geq 2$.

- (b) **What is the distribution of $Y = \sum_{i=1}^3 X_i$? What are the expectation and variance of Y ? Justify your answer.**

Since Y is a linear combination of normally distributed random variables, Y is normally distributed:

$$Y \sim N(\mathbb{E}(Y), \text{Var}(Y))$$

The expectation of Y is:

$$\mathbb{E}(Y) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \mathbb{E}(X_3) = 4 + 1 + 3 = 9$$

The variance of Y is:

$$\text{Var}(Y) = \sum_{i=1}^3 \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$$

Substituting the variances and covariances from the covariance matrix Σ , we get:

$$\text{Var}(Y) = 2 + 0.5 + B + 2(0.5) + 2(-2) = B - \frac{1}{2}$$

Hence, $Y \sim N(9, B - \frac{1}{2})$.

- (c) **Calculate $\mathbb{E}[\exp(X_3) \cdot (X_2 - 1)]$ and justify your answer.**

Since X_2 and X_3 are independent (because the covariance between X_2 and X_3 is zero), we have:

$$\mathbb{E}[\exp(X_3) \cdot (X_2 - 1)] = \mathbb{E}[\exp(X_3)] \cdot \mathbb{E}[X_2 - 1]$$

Now, $\mathbb{E}(X_2 - 1) = \mathbb{E}(X_2) - 1 = 1 - 1 = 0$, so:

$$\mathbb{E}[\exp(X_3) \cdot (X_2 - 1)] = \mathbb{E}[\exp(X_3)] \cdot 0 = 0$$

KLASAUR-2/2

Let X and Y be random variables with the joint probability function

$X \backslash Y$	-1	0	2
$f_{X,Y}(x, y) =$			
0	0.3	0.2	0
1	0.1	0.15	0.05
2	0	0.05	0.15

- (a) Determine the marginal distributions of X and Y .

Solution.

$$f_X(x) = \sum_y f_{X,Y}(x, y)$$

$$f_X(0) = 0.3 + 0.2 + 0 = 0.5$$

$$f_X(1) = 0.1 + 0.15 + 0.05 = 0.3$$

$$f_X(2) = 0 + 0.05 + 0.15 = 0.2$$

$$f_Y(y) = \sum_x f_{X,Y}(x, y)$$

$$f_Y(-1) = 0.3 + 0.1 + 0 = 0.4$$

$$f_Y(0) = 0.2 + 0.15 + 0.05 = 0.4$$

$$f_Y(2) = 0 + 0.05 + 0.15 = 0.2$$

(b) Determine $E(X)$ and $\text{Var}(X)$.

$$E(X) = \sum_x x f_X(x) = 0.7$$

$$E(X^2) = \sum_x x^2 f_X(x) = 1.1$$

$$\text{Var}(X) = E(X^2) - E(X)^2 = 0.61$$

(c) Determine the covariance and correlation between X and Y , and interpret these.

Solution:

$$E(XY) = \sum_x \sum_y xy f_{X,Y}(x, y) = 0.6$$

$$E(Y) = \sum_y y f_Y(y) = 0$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0.6$$

$$E(Y^2) = \sum_y y^2 f_Y(y) = 1.2$$

$$\text{Var}(Y) = E(Y^2) - E(Y)^2 = 1.2$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{0.6}{\sqrt{0.61 \cdot 1.2}} \approx 0.7$$

Interpretation:

The covariance of 0.6 indicates a positive linear relationship between X and Y . The correlation coefficient of approximately 0.7 suggests a moderate positive correlation, meaning that as X increases, Y tends to increase as well, and vice versa.

(d) Determine the conditional distribution of Y given $X = 0$.

Solution:

$$P(Y = y \mid X = 0) = \frac{P(X = 0, Y = y)}{P(X = 0)}$$

$$P(X = 0) = f_X(0) = 0.5$$

y	$P(Y = y \mid X = 0)$
-1	0.6
0	0.4
2	0

KLASAUR-2/3

Let X be Beta distributed with $X \sim \text{Beta}(a, b)$ and Y given X geometrically distributed with $Y | X = x \sim \text{Geom}(x)$.

(a) Calculate $E(Y)$.

Hint:

$$\frac{\Gamma(c+1)}{\Gamma(c)} = c.$$

(b) Calculate the density of $X | Y$.

(c) Calculate $E(X | Y)$.

Solution to Problem 3

Let X be Beta distributed with $X \sim \text{Beta}(a, b)$ and Y given X geometrically distributed with $Y | X = x \sim \text{Geom}(x)$.

(a) **Calculate** $E(Y)$:

$$Y | X \sim \text{Geom}(X)$$

With the law of iterated expectation

$$\begin{aligned} E(Y) &= E[E(Y | X)] = E\left[\frac{1}{X}\right] \\ \mathbb{E}_X\left(\frac{1}{X}\right) &= \int_0^1 \frac{1}{x} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1} dx \\ &= \int_0^1 \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1-1}(1-x)^{b-1} dx \\ &= \frac{\Gamma(a-1)\Gamma(b)}{\Gamma(a)\Gamma(b)} \cdot \frac{\Gamma(a+b)}{\Gamma(a-1+b)} \int_0^1 \frac{\Gamma(a-1+b)}{\Gamma(a-1)\Gamma(b)} x^{a-1-1}(1-x)^{b-1} dx \\ &= \frac{a+b-1}{a-1} \end{aligned}$$

(b) **Calculate the density of** $X | Y$:

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)} \\ &\propto f_{Y|X}(y|x)f_X(x) \\ &= \frac{1}{c} x x^{a-1}(1-x)^{y-1}(1-x)^{b-1} \\ &= \frac{1}{c} x^a(1-x)^{b+y-2} \end{aligned}$$

This is the kernel of a Beta distribution.

$$\implies X|Y = y \sim \text{Beta}(a+1, b+y-1)$$

(c) **Calculate** $E(X | Y)$:

Since $X | Y = y \sim \text{Beta}(a+1, b+y-1)$, the expected value of a Beta distribution $\text{Beta}(\alpha, \beta)$ is given by:

$$E(X | Y = y) = \frac{\alpha}{\alpha + \beta} = \frac{a+1}{a+b+y}$$

KLASAUR-2/4

A fair die is rolled 6000 times independently.

Hint: $\Phi(3.46) \approx 0.9997$

- Use the Central Limit Theorem to approximate the probability that a six is rolled between 900 and 1100 times.
- Use Chebyshev's inequality to determine a lower bound for the probability that a six is rolled between 900 and 1100 times.

Solution

- Central Limit Theorem Approximation:**

Let X be the number of times a six is rolled. X follows a binomial distribution with parameters $n = 6000$ and $p = \frac{1}{6}$, i.e., $X \sim \text{Binomial}(6000, \frac{1}{6})$.

The expected value and variance of X are given by:

$$E(X) = np = 6000 \times \frac{1}{6} = 1000$$

$$\text{Var}(X) = np(1-p) = 6000 \times \frac{1}{6} \times \frac{5}{6} = 833.33$$

Using the Central Limit Theorem, we can approximate the binomial distribution by a normal distribution:

$$X \sim N(1000, 833.33)$$

We want to find $P(900 \leq X \leq 1100)$, so we standardize:

$$Z = \frac{X - 1000}{\sqrt{833.33}} \quad \text{where} \quad Z \sim N(0, 1)$$

For $X = 900$:

$$Z = \frac{900 - 1000}{\sqrt{833.33}} = \frac{-100}{28.87} \approx -3.46$$

For $X = 1100$:

$$Z = \frac{1100 - 1000}{\sqrt{833.33}} = \frac{100}{28.87} \approx 3.46$$

From the standard normal table, we know that:

$$P(Z \leq 3.46) \approx 0.9997$$

Thus,

$$P(900 \leq X \leq 1100) \approx \Phi(3.46) - \Phi(-3.46) = 0.9997 - (1 - 0.9997) = 0.9997 - 0.0003 = 0.9994$$

So, the probability that a six is rolled between 900 and 1100 times is approximately 0.9994.

- Chebyshev's Inequality:**

Chebyshev's inequality states that for any random variable with mean μ and variance σ^2 , the probability that the variable deviates from its mean by more than $k\sigma$ is bounded by:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Here, $\mu = 1000$ and $\sigma = \sqrt{833.33} \approx 28.87$. We want to find the probability that X lies between 900 and 1100, which corresponds to a deviation of $k = \frac{100}{28.87} \approx 3.46$.

Using Chebyshev's inequality:

$$P(|X - 1000| \geq 100) \leq \frac{1}{(3.46)^2} = \frac{1}{11.97} \approx 0.0835$$

Thus, the probability that X lies outside the interval $[900, 1100]$ is at most 0.0835, so the probability that X lies within this interval is at least:

$$P(900 \leq X \leq 1100) \geq 1 - 0.0835 = 0.9165$$

Therefore, the lower bound for the probability is approximately 0.9165.

KLASAUR-2/6

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $U = \{1, 3, 5\}$, $V = \{1, 2, 3\}$, $\mu(A) := \sum_{\omega \in A} \mathbb{I}_U(\omega)$, and $f : \Omega \rightarrow \mathbb{R}$ with

$$f(\omega) = \omega^2.$$

- (a) Specify $\mathcal{F} = \sigma(\{U, V\})$.
- (b) Show that μ is a measure on the measurable space $(\Omega, \mathcal{F})$.
- (c) Compute $\int_V f d\mu$.

Part (c)

We have $\mu(A) = \sum_{\omega \in A} \mathbf{1}_U(\omega) = |A \cap U|$, so $\mu(\{\omega\}) = 1$ if $\omega \in U$ and 0 otherwise. For a finite space the integral is the weighted sum

$$\int_V f d\mu = \sum_{\omega \in V} f(\omega) \mu(\{\omega\}) = \sum_{\omega \in V \cap U} f(\omega).$$

Here $V \cap U = \{1, 3\}$ and $f(\omega) = \omega^2$, hence

$$\int_V f d\mu = 1^2 + 3^2 = 1 + 9 = 10.$$

Answer: $\int_V f d\mu = 10$.

KLASAUR-2/7

Given a continuous random variable X with the density

$$f_X(x) = c \cdot x \cdot I_{[1,3]}(x)$$

- (a) Compute the constant c .

Solution:

The density function must integrate to 1 over the interval $[1, 3]$. Thus, we solve for c :

$$\int_1^3 c \cdot x dx = 1$$

Calculate the integral:

$$\int_1^3 x dx = \left[\frac{x^2}{2} \right]_1^3 = \frac{9}{2} - \frac{1}{2} = 4$$

So we have:

$$c \cdot 4 = 1 \quad \Rightarrow \quad c = \frac{1}{4}$$

Therefore, the constant is $c = \frac{1}{4}$.

(b) Sketch the density.

Solution:

The density is a linear function $f_X(x) = \frac{1}{4} \cdot x$ for $x \in [1, 3]$, which is zero outside this interval. The function starts at $f_X(1) = \frac{1}{4}$ and increases linearly to $f_X(3) = \frac{3}{4}$.

(c) Which of the following statements is correct? Briefly justify (no explicit calculation required).

- The median of X is 2.
- The skewness of X is greater than 0.
- The expectation of X is smaller than the mode of X .

Solution:

- The median of X is not necessarily 2 because the distribution is skewed to the right. Thus, the median is likely closer to 2 but could be slightly lower.
- The skewness is positive because the distribution increases linearly, meaning more mass is concentrated toward the right (higher values of x), which suggests right skewness.
- The mode of X is at the upper boundary, $x = 3$, because the density is increasing. The expectation of X is smaller than the mode because the density increases with x , so the expectation is pulled toward lower values.

(d) Using the transformation of density theorem, calculate the density of the random variable $Y = \frac{1}{1+X}$.

Solution:

The density transformation theorem states that if $Y = g(X)$ and g is a monotonic function, then the density of Y can be found by:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

For $Y = \frac{1}{1+X}$, we have $X = \frac{1}{y} - 1$, so the inverse transformation is $g^{-1}(y) = \frac{1}{y} - 1$. The derivative of $g^{-1}(y)$ with respect to y is:

$$\frac{d}{dy} \left(\frac{1}{y} - 1 \right) = -\frac{1}{y^2}$$

Therefore, the density of Y is:

$$f_Y(y) = f_X \left(\frac{1}{y} - 1 \right) \cdot \left| -\frac{1}{y^2} \right|$$

Since $X \in [1, 3]$, the corresponding range of Y is:

$$Y \in \left[\frac{1}{1+3}, \frac{1}{1+1} \right] = \left[\frac{1}{4}, \frac{1}{2} \right]$$

KLASAUR-2/9

Answer the following questions, each with a brief justification or a calculation with a comprehensible approach.

(a) A and B play the following game: 4 dice are rolled. If at least one six appears, A wins; otherwise, B wins. Is the game fair in the sense that both are equally likely to win on average?

Solution to part (a):

Let X be the event that at least one six is rolled. The probability that no six appears in a single die roll is:

$$P(\text{no six}) = \frac{5}{6}$$

Thus, the probability that no six appears in all 4 rolls is:

$$P(\text{no six in 4 rolls}) = \left(\frac{5}{6}\right)^4 = \frac{625}{1296} \approx 0.48225$$

The probability that at least one six appears (i.e., A wins) is the complement of this:

$$P(\text{at least one six}) = 1 - \left(\frac{5}{6}\right)^4 = 1 - \frac{625}{1296} = \frac{671}{1296} \approx 0.51775$$

Thus, the probability that A wins is approximately 0.51775, and the probability that B wins is approximately 0.48225. Since these probabilities are not equal, the game is not fair, as A is more likely to win than B.

- (b) In a population, 5 percent of people suffer from high blood pressure. Of these 5 percent, 75% regularly drink alcohol. It is also known that 50% of people without high blood pressure regularly drink alcohol. What percentage of regular alcohol drinkers suffer from high blood pressure?

Solution to part (b):

We are given the following information:

- $P(\text{high blood pressure}) = 0.05$
- $P(\text{regular alcohol consumption} \mid \text{high blood pressure}) = 0.75$
- $P(\text{regular alcohol consumption} \mid \text{no high blood pressure}) = 0.50$

We want to find the percentage of regular alcohol drinkers who suffer from high blood pressure, i.e., $P(\text{high blood pressure} \mid \text{regular alcohol consumption})$. Using Bayes' theorem:

$$\begin{aligned} &P(\text{high blood pressure} \mid \text{regular alcohol consumption}) \\ &= \frac{P(\text{regular alcohol consumption} \mid \text{high blood pressure}) \cdot P(\text{high blood pressure})}{P(\text{regular alcohol consumption})} \end{aligned}$$

First, we need to calculate $P(\text{regular alcohol consumption})$:

$$P(\text{regular alcohol consumption}) = P(\text{regular alcohol consumption} \mid \text{high blood pressure}) \cdot P(\text{high blood pressure}) + P(\text{regular alcohol consumption} \mid \text{no high blood pressure}) \cdot P(\text{no high blood pressure})$$

$$P(\text{regular alcohol consumption}) = 0.75 \cdot 0.05 + 0.50 \cdot 0.95 = 0.0375 + 0.475 = 0.5125$$

Now, we can calculate $P(\text{high blood pressure} \mid \text{regular alcohol consumption})$:

$$P(\text{high blood pressure} \mid \text{regular alcohol consumption}) = \frac{0.75 \cdot 0.05}{0.5125} = \frac{0.0375}{0.5125} \approx 0.0732$$

Thus, approximately 7.32% of regular alcohol consumers suffer from high blood pressure.

- (c) Let X be a continuous random variable with cumulative distribution function F_X and a 0.25-quantile of 3. Which of the following statements is/are true?

- (i) $F_X(3) = 0.25$ (TRUE. See definition page 169)
- (ii) $F_X(0.25) = 3$ (FALSE)
- (iii) $F_X^{-1}(3) = 0.25$ (FALSE)

- (d) Let Y be a discrete random variable with support $T_Y = \mathbb{N}$ and cumulative distribution function F_Y such that $F_Y(33/10) = 0.5$ and $F_Y(2) \neq F_Y(3) \neq F_Y(4)$. Determine which of the following statements follow from this information:
- (i) The median of Y is 3.
 - (ii) $P(Y < 3) \leq 0.5$
 - (iii) The expected value of Y is 3.

Solution to part (d):

We are given the following information about the random variable Y :

- The cumulative distribution function F_Y satisfies $F_Y\left(\frac{33}{10}\right) = 0.5$.
- $F_Y(2) \neq F_Y(3) \neq F_Y(4)$, meaning that the probabilities are distinct for $Y = 2$, $Y = 3$, and $Y = 4$.

Let's evaluate each statement:

- (i) **The median of Y is 3.**

By definition, the median is the value such that $P(Y \leq \text{median}) = 0.5$. We know that $F_Y(33/10) = 0.5$. Since $F_Y(33/10) \approx F_Y(3)$ and $F_Y(2) \neq F_Y(3) \neq F_Y(4)$, it is reasonable to conclude that the median of Y is indeed 3. Therefore, this statement is **true**.

- (ii) $P(Y < 3) \leq 0.5$.

Since $F_Y(33/10) = 0.5$, we know that $F_Y(3) \geq 0.5$. Therefore, $P(Y < 3) = F_Y(2)$ must be less than or equal to 0.5. This statement is **true**.

- (iii) **The expected value of Y is 3.**

There is not enough information provided to directly calculate the expected value of Y . The quantile information tells us about the distribution's shape but not enough to determine the expected value. Therefore, this statement is **false**.

Thus, the correct statements are (i) and (ii).

- (e) What is the distribution of the random variable $Z = 3U - 12$ if $U \sim \mathcal{N}(\mu = 4, \sigma^2 = 5)$?

Solution to part (e):

We are given that $U \sim \mathcal{N}(4, 5)$, meaning that U is normally distributed with mean $\mu = 4$ and variance $\sigma^2 = 5$. The transformation applied to U is $Z = 3U - 12$.

The mean and variance of Z can be derived from the transformation of U .

• **Mean of Z :**

The mean of a linear transformation $Z = aU + b$ of a normally distributed variable U is given by:

$$E(Z) = aE(U) + b$$

Substituting $a = 3$, $b = -12$, and $E(U) = 4$, we get:

$$E(Z) = 3 \times 4 - 12 = 12 - 12 = 0$$

• **Variance of Z :**

The variance of a linear transformation $Z = aU + b$ of a normally distributed variable U is given by:

$$\text{Var}(Z) = a^2 \text{Var}(U)$$

Substituting $a = 3$ and $\text{Var}(U) = 5$, we get:

$$\text{Var}(Z) = 3^2 \times 5 = 9 \times 5 = 45$$

Thus, Z is normally distributed with mean 0 and variance 45, i.e.,

- (f) Assume a bottle return machine accepts each bottle presented to it with a probability $p < 1$. Let F be the number of bottles that need to be presented to the machine in order to get a receipt for m accepted bottles. Using a known parametric distribution from the

lecture, describe F and provide the parameter values. What additional assumptions about the data-generating process must you make?

Solution to part (f):

The problem describes a scenario where a machine accepts bottles with a probability $p < 1$, and we need to find the distribution of the number of bottles presented to the machine in order to get m accepted bottles.

This is a typical setup for a **Negative Binomial Distribution**. The negative binomial distribution models the number of trials needed to achieve a fixed number of successes in independent Bernoulli trials. In this case, the "success" is the machine accepting a bottle, and the "trials" are the number of bottles presented to the machine.

Thus, F , the number of bottles that need to be presented to get m accepted bottles, follows a negative binomial distribution with parameters:

$$F \sim \text{NegBin}(m, p)$$

where m is the number of accepted bottles (the number of "successes"), and p is the probability that a bottle is accepted (the probability of "success").

Parameter values:

- m : the number of accepted bottles (successes) required to get the receipt.
- p : the probability that a bottle is accepted.

Assumptions:

- The acceptance of each bottle is independent of the others (independent trials).
- The probability of acceptance p is constant for each bottle.

These assumptions are necessary to apply the negative binomial distribution to model the process.

- (g) The following mosaic plot shows the observed relationship between the variables gender (male/female) and exam success (passed/failed) for a statistics exam.
- (i) For which gender is the failure rate higher?
 - (ii) Are there more men or women who pass the exam?
 - (iii) In addition to gender and exam success, suppose we have additional data on the variable "field of study" with two levels: "minor" and "major." This variable is empirically independent of gender and exam success. Half of the examinees are minor students, and the other half are major students. Sketch a mosaic plot showing the joint distribution of these three variables. (Only a schematic drawing is required, no precise drawing needed.)

Solution to part (g):

- (i) **For which gender is the failure rate higher?**

The mosaic plot shows that the proportion of men who did not pass the exam is higher than the proportion of women who did not pass. Therefore, the failure rate is higher for men.

- (ii) **Are there more men or women who pass the exam?**

From the mosaic plot, we can see that the width of the "passed" section for men is larger than that for women. This indicates that there are more men than women who pass the exam.

- (iii) **Schematic drawing of a mosaic plot for the joint distribution of gender, exam success, and field of study:**

In this case, we need to add a third variable "field of study" (minor/major) to the plot, which is independent of both gender and exam success. This means that the "minor" and "major" sections will be equally split for both men and women, as well as for those who passed or did not pass.

The mosaic plot will consist of four quadrants for each gender and exam success combination. Each quadrant will then be split into two equal sections, one for "minor" students and one for "major" students.

Since "field of study" is independent of the other variables, the size of the sections for "minor" and "major" will be the same within each quadrant.

- (h) There are a large number n of independent Poisson-distributed random variables, each with the same rate λ . How is the sum of these random variables distributed exactly, and which distribution does this sum approximate?

Solution to part (h):

Let $X_1, X_2, \dots, X_n$ be independent Poisson-distributed random variables with rate λ , i.e., $X_i \sim \text{Poisson}(\lambda)$ for all $i = 1, 2, \dots, n$.

• **Exact Distribution:**

The sum of independent Poisson random variables is also Poisson-distributed. Therefore, the sum of these n Poisson random variables is given by:

$$S_n = X_1 + X_2 + \dots + X_n$$

where $S_n \sim \text{Poisson}(n\lambda)$.

• **Approximate Distribution:**

When n is large, by the Central Limit Theorem (CLT), the sum of a large number of Poisson random variables can be approximated by a normal distribution. Specifically, S_n can be approximated by a normal distribution with mean $n\lambda$ and variance $n\lambda$:

$$S_n \approx \mathcal{N}(n\lambda, n\lambda)$$

Thus, the exact distribution of the sum is $\text{Poisson}(n\lambda)$, and for large n , this sum can be approximated by a normal distribution $\mathcal{N}(n\lambda, n\lambda)$.

KLASAUR-3/1

A study examined the relationships between the continuation of marriage after seven years ("Divorced" / "Not Divorced"), the division of household chores between the partners ("Fair" / "Unfair"), and the income differences between the partners ("Man earns more" / "Man earns less") in 1000 heterosexual couples. In total, 200 of the 1000 couples were already divorced after seven years.

Hint: Fictional data, invented numbers. In the following, no psychological or sociological interpretations are asked, but a direct interpretation of the calculated measures.

- (a) Complete the marginal joint frequencies of the continuation of marriage and the division of household chores without considering income differences (upper table) as well as the joint frequency distribution of all three variables (lower two tables) in the following contingency tables.

	Divorced	Not Divorced	Σ
Fair	80	420	500
Unfair	120	380	500
Σ	200	800	1000

- (b) The researchers are primarily interested in possible differences in divorce rates between couples where household chores are divided fairly and couples where household chores are divided unfairly. Calculate the corresponding odds ratio and briefly interpret your result.

Solution for (b):

The odds ratio is given by:

$$\text{OR} = \frac{\left(\frac{\text{Divorced Fair}}{\text{Not Divorced Fair}} \right)}{\left(\frac{\text{Divorced Unfair}}{\text{Not Divorced Unfair}} \right)} = \frac{\frac{80}{420}}{\frac{120}{380}} = \frac{80 \times 380}{420 \times 120} = 0.6048$$

Interpretation: The odds of getting divorced for couples who divide household chores unfairly are about 60

- (c) Based on the data from the study, are there indications that the relationship between the division of household chores and the continuation of marriage is modified by the income differences between the partners? Calculate the relevant odds ratios and interpret your result. Name the term for this phenomenon.

Solution for (c):

To determine whether income differences between the partners modify the relationship between the division of household chores and the continuation of the marriage, we calculate the odds ratios separately for both income groups.

For the "Man earns more" group:

$$\text{OR}_{\text{man earns more}} = \frac{\frac{20}{60}}{\frac{60}{90}} = \frac{20 \times 90}{60 \times 60} = 0.5$$

For the "Man earns less" group:

$$\text{OR}_{\text{man earns less}} = \frac{\frac{60}{40}}{\frac{40}{100}} = \frac{60 \times 100}{40 \times 40} = 3.75$$

Interpretation: There is evidence that the effect of fair or unfair division of household chores on divorce is modified by income differences. Specifically, when the man earns less, the odds of divorce are higher when chores are unfairly divided, compared to when the man earns more.

The term for this phenomenon is called **interaction** or **effect modification**.

KLASAUR-3/4

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $U = \{1, 2, 3, 4\}$, $V = \{3, 4, 5, 6\}$, $\mathcal{E} = \{U, V\}$, and $f : \Omega \rightarrow \mathbb{R}$ with $f(\omega) = \omega$. Let μ_Z be the counting measure.

- The sigma-algebra $\sigma(\mathcal{E})$ is generated over Ω . Determine $\sigma(\mathcal{E})$.
- Compute $\int_V f d\mu_Z$.
- Specify the null set of the measure μ_Z on the sigma-algebra generated by Ω .

Solution for (a):

Atoms from U and V : $U \cap V = \{3, 4\}$, $U \cap V^c = \{1, 2\}$, $U^c \cap V = \{5, 6\}$. (And $U^c \cap V^c = \emptyset$.) Hence $\sigma(\mathcal{E})$ is all unions of $\{1, 2\}$, $\{3, 4\}$, $\{5, 6\}$: $\sigma(\mathcal{E}) = \{\emptyset, \{1, 2\}, \{3, 4\}, \{5, 6\}, U, V, \{1, 2, 5, 6\}, \Omega\}$.

Solution for (b):

Counting measure $\Rightarrow$ integrate by summing over V :

$$\int_V f d\mu_Z = \sum_{\omega \in V} f(\omega) = 3 + 4 + 5 + 6 = 18.$$

Solution for (c):

With counting measure, the only set of measure 0 is the empty set:

$$\mathcal{N}(\mu_Z) = \{\emptyset\}.$$

KLASAUR-3/5

Given a continuous random variable X with density function:

$$f_X(x) = c \cdot x^2 \cdot \mathbb{I}_{[0,3]}(x)$$

(a) Show that $c = \frac{1}{9}$.

To determine c , we use the fact that the total integral of a probability density function must equal 1:

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

Since $f_X(x) = c \cdot x^2$ only for $x \in [0, 3]$, the integral simplifies to:

$$\int_0^3 c \cdot x^2 dx = 1$$

Now, compute the integral:

$$\begin{aligned} c \int_0^3 x^2 dx &= 1 \\ c \left[\frac{x^3}{3} \right]_0^3 &= 1 \\ c \left(\frac{3^3}{3} - 0 \right) &= 1 \\ c \cdot 9 &= 1 \\ c &= \frac{1}{9} \end{aligned}$$

Thus, $c = \frac{1}{9}$.

(b) Sketch the density function

The density function $f_X(x) = \frac{1}{9}x^2$ is a parabola that is defined only in the interval $[0, 3]$. Outside this interval, the function is zero. The maximum value of the density occurs at $x = 3$.

(c) Calculate the expected value of X .

The expected value of a continuous random variable is given by:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Since $f_X(x) = \frac{1}{9}x^2$ for $x \in [0, 3]$, this becomes:

$$\mathbb{E}[X] = \int_0^3 x \cdot \frac{1}{9}x^2 dx = \frac{1}{9} \int_0^3 x^3 dx$$

Now, compute the integral:

$$\begin{aligned} \frac{1}{9} \left[\frac{x^4}{4} \right]_0^3 &= \frac{1}{9} \cdot \frac{3^4}{4} \\ \mathbb{E}[X] &= \frac{1}{9} \cdot \frac{81}{4} = \frac{9}{4} = 2.25 \end{aligned}$$

Thus, the expected value of X is 2.25.

(d) Calculate the Median of X

The median m of a continuous random variable X is the value such that:

$$P(X \leq m) = \frac{1}{2}$$

This means that the integral of the probability density function from 0 to m must be $\frac{1}{2}$, i.e.,

$$\int_0^m f_X(x) dx = \frac{1}{2}$$

Using the density function $f_X(x) = \frac{1}{9}x^2$ for $x \in [0, 3]$, we need to solve:

$$\int_0^m \frac{1}{9}x^2 dx = \frac{1}{2}$$

First, compute the integral:

$$\frac{1}{9} \left[\frac{x^3}{3} \right]_0^m = \frac{1}{9} \cdot \frac{m^3}{3}$$

This gives us:

$$\frac{m^3}{27} = \frac{1}{2}$$

Multiply both sides by 27:

$$m^3 = \frac{27}{2}$$

Taking the cube root:

$$m = \sqrt[3]{\frac{27}{2}} = 3 \cdot \sqrt[3]{\frac{1}{2}} = 3 \cdot 2^{-\frac{1}{3}}$$

Thus, the median is:

$$m = 3 \cdot 2^{-\frac{1}{3}} \approx 2.38$$

(e) Calculate the density of $Y = \exp(X + 1)$ using the transformation theorem.

We are given that $Y = \exp(X + 1)$ where X has a known density function:

$$f_X(x) = \frac{1}{9}x^2 \cdot \mathbb{I}_{[0,3]}(x)$$

We need to find the density of Y using the transformation theorem. The general formula for transforming a density is:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

Step 1: Find the inverse of the transformation

We have:

$$Y = \exp(X + 1)$$

Solving for X :

$$X = \ln(Y) - 1$$

Thus, $g^{-1}(Y) = \ln(Y) - 1$.

Step 2: Compute the derivative

The derivative of $g^{-1}(y) = \ln(y) - 1$ is:

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{y}$$

Step 3: Apply the transformation theorem

Substituting into the formula for $f_Y(y)$:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{1}{y} \right|$$

Substituting $g^{-1}(y) = \ln(y) - 1$ into the density function $f_X(x) = \frac{1}{9}x^2$:

$$f_X(g^{-1}(y)) = f_X(\ln(y) - 1) = \frac{1}{9}(\ln(y) - 1)^2 \cdot \mathbb{I}_{[\exp(1), \exp(4)]}(y)$$

Now, multiply by $\left| \frac{1}{y} \right|$:

$$f_Y(y) = \frac{1}{9}(\ln(y) - 1)^2 \cdot \frac{1}{y} \cdot \mathbb{I}_{[\exp(1), \exp(4)]}(y)$$

Final Answer

The density function of the random variable $Y = \exp(X + 1)$ is:

$$f_Y(y) = \frac{1}{9} \cdot \frac{(\ln(y) - 1)^2}{y} \cdot \mathbb{I}_{[\exp(1), \exp(4)]}(y)$$

KLASAUR-3/6

Let $\mathbf{X} = (X_1, X_2)$ be multivariate normally distributed with $\mu = (0, 2)$ and $\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$.

- (a) Give the distributions of X_1 and X_2 with their expected value and variance.
- (b) Are X_1 and X_2 stochastically independent? Justify.
- (c) Calculate the probability $P(X_1 > 1, X_2 \leq 1)$. Round your result to 3 decimal places.
Hint: $\Phi(0.5) = 0.6915, \Phi(1) = 0.8413, \Phi(2) = 0.97725$
- (d) Estimate the lower bound for $\mathbb{E}(|X_1| + |X_2|)$.
- (e) Are the following matrices valid covariance matrices? Justify.

$$\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0.5 & 0 \\ 0.5 & 2 & -2 \\ 0 & -2 & -1 \end{pmatrix}$$

- (a) The distribution of X_1 is $N(0, 4)$, i.e., normally distributed with a mean of 0 and a variance of 4. Similarly, the distribution of X_2 is $N(2, 1)$, i.e., normally distributed with a mean of 2 and a variance of 1.
- (b) Since the covariance matrix Σ is diagonal, X_1 and X_2 are independent. The off-diagonal elements in the covariance matrix represent the covariances between X_1 and X_2 , and since these are zero, X_1 and X_2 are stochastically independent.
- (c) The probability $P(X_1 > 1, X_2 \leq 1)$ can be calculated as:

$$P(X_1 > 1, X_2 \leq 1) = P(X_1 > 1) \cdot P(X_2 \leq 1).$$

First, we calculate $P(X_1 > 1)$. Since $X_1 \sim N(0, 4)$, we standardize X_1 :

$$Z_1 = \frac{X_1 - 0}{\sqrt{4}} = \frac{X_1}{2}.$$

Therefore, $P(X_1 > 1) = P(Z_1 > \frac{1}{2}) = 1 - \Phi(0.5) = 1 - 0.6915 = 0.3085$.

Now, calculate $P(X_2 \leq 1)$. Since $X_2 \sim N(2, 1)$, we standardize X_2 :

$$Z_2 = \frac{X_2 - 2}{\sqrt{1}} = X_2 - 2.$$

Therefore, $P(X_2 \leq 1) = P(Z_2 \leq -1) = \Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$.

Thus, the required probability is:

$$P(X_1 > 1, X_2 \leq 1) = 0.3085 \times 0.1587 = 0.04895.$$

Rounded to three decimal places, this is 0.049.

- (d) Since $x \mapsto |x|$ is convex, by Jensen's inequality we have

$$\mathbb{E}|X_i| \geq |\mathbb{E}X_i|, \quad i = 1, 2.$$

Therefore,

$$\mathbb{E}(|X_1| + |X_2|) = \mathbb{E}|X_1| + \mathbb{E}|X_2| \geq |\mathbb{E}X_1| + |\mathbb{E}X_2|.$$

From the mean vector $\mu = (0, 2)$ we know that $\mathbb{E}X_1 = 0$ and $\mathbb{E}X_2 = 2$. Hence

$$\mathbb{E}(|X_1| + |X_2|) \geq 0 + 2 = 2.$$

- (e) A matrix is a valid covariance matrix if it is symmetric and positive semi-definite. We analyze the two matrices:

$$\text{For } \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}:$$

$$\text{Determinant} = 1 \cdot 1 - 3 \cdot 3 = 1 - 9 = -8.$$

Since the determinant is negative, this matrix is not positive semi-definite and thus not a valid covariance matrix.

$$\text{For } \begin{pmatrix} 3 & 0.5 & 0 \\ 0.5 & 2 & -2 \\ 0 & -2 & -1 \end{pmatrix}: \text{ We check whether all the leading principal minors are non-}$$

negative. If they are, the matrix is positive semi-definite. After calculating, the matrix is not positive semi-definite and thus not a valid covariance matrix.

KLASAUR-3/7

- (a) In a population, 2% suffer from a disease. Of these two percent, 80% smoke regularly. It is also known that 30% of people who do not have the disease smoke regularly. How many percent of regular smokers suffer from the disease? Round your result to 3 decimal places.
- (b) Let X given Y be geometrically distributed with $X|Y = y \sim \text{Geom}(y)$, and let Y be uniformly distributed with $Y \sim U(1, 2)$. Compute $\mathbb{E}(X)$.

Solution

(a) Percentage of Smokers with the Disease

We use **Bayes' theorem** to find the probability that a regular smoker has the disease.

Let:

- D denote having the disease,
- S denote smoking regularly.

We are given the following probabilities:

$$P(D) = 0.02, \quad P(S|D) = 0.80, \quad P(S|D^c) = 0.30,$$

where D^c represents not having the disease.

We want to find $P(D|S)$, the probability of having the disease given that the person smokes regularly. By Bayes' theorem:

$$P(D|S) = \frac{P(S|D)P(D)}{P(S)}.$$

First, calculate $P(S)$, the probability of smoking regularly:

$$P(S) = P(S|D)P(D) + P(S|D^c)P(D^c).$$

Substitute the known values:

$$P(S) = 0.80 \times 0.02 + 0.30 \times (1 - 0.02) = 0.016 + 0.30 \times 0.98 = 0.016 + 0.294 = 0.31.$$

Now, use Bayes' theorem to find $P(D|S)$:

$$P(D|S) = \frac{0.80 \times 0.02}{0.31} = \frac{0.016}{0.31} \approx 0.0516.$$

Thus, about 5.16% of regular smokers suffer from the disease.

(b) Expected Value of X

Given:

$$X|Y = y \sim \text{Geom}(y), \quad Y \sim U(1, 2),$$

we know the expected value of a geometric distribution is:

$$\mathbb{E}(X|Y = y) = \frac{1}{y}.$$

To find $\mathbb{E}(X)$, we use the law of total expectation:

$$\mathbb{E}(X) = \mathbb{E}[\mathbb{E}(X|Y)] = \mathbb{E}\left[\frac{1}{Y}\right].$$

Since $Y \sim U(1, 2)$, the expected value of $\frac{1}{Y}$ is given by:

$$\mathbb{E}\left[\frac{1}{Y}\right] = \int_1^2 \frac{1}{y} f_Y(y) dy = \int_1^2 \frac{1}{y} \cdot 1 dy.$$

This integral evaluates to:

$$\mathbb{E}\left[\frac{1}{Y}\right] = \ln(2) - \ln(1) = \ln(2).$$

Thus, the expected value of X is:

$$\mathbb{E}(X) = \ln(2) \approx 0.693.$$

KLASAUR-3/8

- (a) A person finds that they pass each exam with a 20% probability without studying. How many exams, on average, would they need to take to pass one? What is the resulting variance?
- (b) Lufthansa has observed that about 18% of passengers do not show up for their booked flight. To maximize the occupancy of planes, they sell more than the available 150 seats on an Airbus A320. Use the central limit theorem to calculate the probability that more than 150 passengers show up for a flight when 170 tickets are sold. Round your result to 3 decimal places.

Hint: $\Phi(2.116) = 0.983$

Solution

(a) Expected Number of Exams and Variance

The number of exams the person must take to pass can be modeled by a geometric distribution with success probability $p = 0.20$.

The expected number of trials (exams) to achieve the first success is given by the expected value of a geometric random variable:

$$\mathbb{E}(X) = \frac{1}{p} = \frac{1}{0.20} = 5.$$

Thus, on average, the person would need to take 5 exams to pass.

The variance of a geometric random variable is given by:

$$\text{Var}(X) = \frac{1-p}{p^2}.$$

Substitute $p = 0.20$:

$$\text{Var}(X) = \frac{1-0.20}{(0.20)^2} = \frac{0.80}{0.04} = 20.$$

Thus, the variance is 20.

(b) Probability That More Than 150 Passengers Show Up

Let X denote the number of passengers who show up for a flight. X follows a binomial distribution with parameters $n = 170$ and $p = 0.82$ (since 18% of passengers do not show up, the probability that a passenger shows up is $1 - 0.18 = 0.82$).

The expected value and variance of X are:

$$\mathbb{E}(X) = np = 170 \times 0.82 = 139.4,$$

$$\text{Var}(X) = np(1-p) = 170 \times 0.82 \times 0.18 = 25.164.$$

By the central limit theorem, for large n , X can be approximated by a normal distribution:

$$X \sim N(\mathbb{E}(X), \text{Var}(X)) = N(139.4, 25.164).$$

We want to find $P(X > 150)$. First, we standardize:

$$Z = \frac{X - 139.4}{\sqrt{25.164}} = \frac{150 - 139.4}{5.016} = 2.116.$$

From the standard normal table, we know that:

$$P(Z > 2.116) = 1 - \Phi(2.116) = 1 - 0.983 = 0.017.$$

Thus, the probability that more than 150 passengers show up is approximately 0.017.

BLATT 3/1

Let $\mathbb{P}$ be a probability measure (i.e., a normalized measure with $\mathbb{P}(\Omega) = 1$) on the measurable space $(\Omega, \mathcal{F})$, where $A, B \in \mathcal{F}$.

- (a) If $\mathbb{P}(A) = \frac{1}{3}$ and $\mathbb{P}(\bar{B}) = \frac{1}{4}$, can A and B then be disjoint?
- (b) Prove or disprove: $\mathbb{P}(A) = \mathbb{P}(\bar{B}) \Rightarrow \bar{A} = B$.
- (c) Prove or disprove: $\mathbb{P}(B) = 0 \Rightarrow \mathbb{P}(A \cap B) = 0$.
- (d) Let $\Omega = \{i \mid i \in \mathbb{N}_0\}$ with elementary events $\omega_i = i$. Furthermore, let $\mathbb{P}(\{\omega_i\}) = \frac{c}{i!}$. What is the value of c ?

Solution:

$$(a) \mathbb{P}(A) = \frac{1}{3}, \mathbb{P}(B) = 1 - \mathbb{P}(\bar{B}) = \frac{3}{4}$$

$$A \cap B = \emptyset \Rightarrow \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) = \frac{1}{3} + \frac{3}{4} = \frac{13}{12} > 1 \quad \text{Contradiction}$$

$$(b) \mathbb{P}(A) = \mathbb{P}(\bar{B}) \Rightarrow \bar{A} = B?$$

Counterexample:

$$\Omega = \{1, 2, \dots, 6\}, \quad A = \{1, 2\}, \quad \bar{B} = \{5, 6\} \quad (\text{Dice rolling})$$

$$\mathbb{P}(A) = \frac{2}{6} = \mathbb{P}(\bar{B}), \quad \text{aber} \quad \bar{A} = \{3, 4, 5, 6\} \neq B = \{1, 2, 3, 4\}$$

$$(c) B = (B \cap A) \cup (B \cap \bar{A})$$

$$\Rightarrow \mathbb{P}(B) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap \bar{A}) = 0 \quad (\geq 0) \Rightarrow \mathbb{P}(B \cap A) = 0$$

$$(d)$$

$$\sum_{i=0}^{\infty} \mathbb{P}(\omega_i) = 1$$

$$\sum_{i=0}^{\infty} \frac{c}{i!} = c \sum_{i=0}^{\infty} \frac{1}{i!} = c \cdot e^1 \Rightarrow c = e^{-1}$$

BLATT 3/2

(a) Give Proof: Given events A, B, C . If A, B, C are independent, then $A \cup B$ and C are also independent.

Solution.

$$\mathbb{P}(A) \cdot \mathbb{P}(B) = \mathbb{P}(A \cap B)$$

$$\mathbb{P}(A) \cdot \mathbb{P}(C) = \mathbb{P}(A \cap C)$$

$$\mathbb{P}(B) \cdot \mathbb{P}(C) = \mathbb{P}(B \cap C)$$

To show:

$$\mathbb{P}(A \cup B) \cdot \mathbb{P}(C) = \mathbb{P}((A \cup B) \cap C)$$

$$\begin{aligned} \mathbb{P}((A \cup B) \cap C) &= \mathbb{P}((A \cap C) \cup (B \cap C)) \\ &= \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C) \\ &= \mathbb{P}(A) \cdot \mathbb{P}(C) + \mathbb{P}(B) \cdot \mathbb{P}(C) - \mathbb{P}(A) \cdot \mathbb{P}(B) \cdot \mathbb{P}(C) \\ &= (\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A) \cdot \mathbb{P}(B)) \cdot \mathbb{P}(C) \\ &= \mathbb{P}(A \cup B) \cdot \mathbb{P}(C) \end{aligned}$$

(b) For two sets $\mathcal{E}_1$ and $\mathcal{E}_2$ of events in a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ with $\mathcal{E}_1 \subseteq \mathcal{E}_2$, show that: $\mathcal{E}_1$ and $\mathcal{E}_2$ are independent if and only if $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$ for all $A \in \mathcal{E}_1$.

Solution.

Step 1: Given two sets of events $\mathcal{E}_1 \subseteq \mathcal{E}_2$:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \quad \text{for all } A \in \mathcal{E}_1$$

This implies:

$$\mathbb{P}(A) = 1 \quad \text{or} \quad \mathbb{P}(A) = 0$$

$$\mathbb{P}(A \cap \mathcal{E}_2) = \mathbb{P}(A)\mathbb{P}(\mathcal{E}_2)$$

$$\mathbb{P}(A \cap \mathcal{E}_2) = \mathbb{P}(A)\mathbb{P}(\mathcal{E}_2)$$

$$\mathbb{P}(A \cap \mathcal{E}_2) = \mathbb{P}(A) \quad \text{if and only if} \quad \mathbb{P}(A) = 1 \quad \text{or} \quad \mathbb{P}(A) = 0$$

Step 2:

For $\mathbb{P}(A) = 1$ and $\mathbb{P}(A) = 0$:

$$\mathbb{P}(A \cap \mathcal{E}_2) = \mathbb{P}(A)\mathbb{P}(\mathcal{E}_2)$$

$$\mathcal{E}_1 \subseteq \mathcal{E}_2 \implies \mathbb{P}(A) = 1 \implies \mathbb{P}(\mathcal{E}_2) = 1 \implies \mathbb{P}(A \cap B) = 1$$

$$\mathcal{E}_1 \subseteq \mathcal{E}_2 \implies \mathbb{P}(A) = 0 \implies \mathbb{P}(\mathcal{E}_2) = 0 \implies \mathbb{P}(A \cap B) = 0$$

$$\mathbb{P}(A \cap B) = \begin{cases} 1 & \text{if } \mathbb{P}(A) = 1 \\ 0 & \text{if } \mathbb{P}(A) = 0 \end{cases}$$

$$\mathbb{P}(A)\mathbb{P}(B) = \begin{cases} 1 & \text{if } \mathbb{P}(A) = 1 \\ 0 & \text{if } \mathbb{P}(A) = 0 \end{cases}$$

BLATT 3/3

A random variable X is called (discretely) uniformly distributed over $\{1, \dots, n\}$, $n \in \mathbb{N}$, if its probability mass function satisfies $\mathbb{P}(X = j) = \frac{1}{n}$ for all $j = 1, \dots, n$. The random variables X and Y are now assumed to be independent and uniformly distributed over $\{1, \dots, n\}$. Determine the probability mass function of the random variable $Z = X + Y$, i.e., the individual probabilities $\mathbb{P}(Z = j)$ for all j in the range of Z .

Hint: Two discrete random variables X and Y are independent if the events $\{X = k\}$ and $\{Y = j\}$ are independent events.

Solution:

Assumption:

$$\mathbb{P}(X = i) = \mathbb{P}(Y = i) = \begin{cases} \frac{1}{n}, & i = 1, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

The support of Z is $C = \{2, \dots, 2n\}$, i.e., $\mathbb{P}(Z \in C) = 1$. We now distinguish between two cases:

- Let $2 \leq j \leq n$. Then:

$$\begin{aligned} \mathbb{P}(Z = j) &= \mathbb{P}(X + Y = j) = \sum_{k=1}^{j-1} \mathbb{P}(X = k, Y = j - k) \stackrel{\text{indep.}}{=} \sum_{k=1}^{j-1} \mathbb{P}(X = k) \mathbb{P}(Y = j - k) \\ &= \sum_{k=1}^{j-1} \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^{j-1} 1 = \frac{j-1}{n^2}. \end{aligned}$$

- Let $n < j \leq 2n$. Then:

$$\begin{aligned} \mathbb{P}(Z = j) &= \mathbb{P}(X + Y = j) = \sum_{k=j-n}^n \mathbb{P}(X = k, Y = j - k) \stackrel{\text{indep.}}{=} \sum_{k=j-n}^n \mathbb{P}(X = k) \mathbb{P}(Y = j - k) \\ &= \sum_{k=j-n}^n \frac{1}{n^2} = \frac{1}{n^2} \sum_{k=j-n}^n 1 = \frac{2n - j + 1}{n^2}. \end{aligned}$$

Here, we have used the fact that $\mathbb{P}(Y = j - k) = 0$ if and only if $k \leq j - n - 1$.

BLATT 4/1

Let $\Omega_1 = \Omega_2 = \{1, 2, 3, 4, 5, 6\}$ each with a σ -algebra $\mathcal{F}_1 = \mathcal{F}_2 = \sigma(\mathcal{E})$, where

$$\mathcal{E} = \{\{1, 2\}, \{3, 4\}\}.$$

Consider the function $f(x) = \max(1, x - 1)$.

- Explicitly provide the two set systems $f^{-1}(\mathcal{F}_2)$ and $\{B \mid f^{-1}(B) \in \mathcal{F}_1\}$ and show that they are σ -algebras.
- Is f measurable with respect to $\mathcal{F}_1 - \mathcal{F}_2$?
- Show that $f(\mathcal{F}_1)$ is not a σ -algebra. (Which two properties are not satisfied?)

Solution.

(a) $f(x) = \max(1, x - 1)$

$$x = 1 : f(1) = \max(1, 0) = 1$$

$$x = 2 : f(2) = \max(1, 1) = 1$$

$$x = 3 : f(3) = \max(1, 2) = 2$$

$$x = 4 : f(4) = \max(1, 3) = 3$$

$$x = 5 : f(5) = \max(1, 4) = 4$$

$$x = 6 : f(6) = \max(1, 5) = 5$$

$\mathcal{F}_2$ is the σ -algebra generated by $\mathcal{E}$, where $\mathcal{E} = \{\{1, 2\}, \{3, 4\}\}$.

$$\mathcal{F}_1 = \mathcal{F}_2 = \{\emptyset, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{1, 2, 3, 4\}, \{1, 2, 5, 6\}, \{3, 4, 5, 6\}, \Omega_2 = \{1, 2, 3, 4, 5, 6\}\}$$

$$f^{-1}(\mathcal{F}_2) = \{\emptyset, \{1, 2, 3\}, \{4, 5\}, \{6\}, \{1, 2, 3, 4, 5\}, \{1, 2, 3, 6\}, \{4, 5, 6\}, \{1, 2, 3, 4, 5, 6\}\}$$

$$\{B \mid f^{-1}(B) \in \mathcal{F}_1\} = M_1 \cup M_2 \cup \dots \cup M_8 \text{ with}$$

$$M_1 = \{B \mid f^{-1}(B) = \emptyset\} = \{\emptyset, \{6\}\}$$

$$M_2 = \{B \mid f^{-1}(B) = \{1, 2\}\} = \{\{1\}, \{1, 6\}\}$$

$$M_3 = \{B \mid f^{-1}(B) = \{3, 4\}\} = \{\{2, 3\}, \{2, 3, 6\}\}$$

$$M_4 = \{B \mid f^{-1}(B) = \{5, 6\}\} = \{\{4, 5\}, \{4, 5, 6\}\}$$

$$M_5 = \{B \mid f^{-1}(B) = \{1, 2, 3\}\} = \{\{1, 2, 3\}, \{1, 2, 3, 6\}\}$$

$$M_6 = \{B \mid f^{-1}(B) = \{1, 2, 5, 6\}\} = \{\{1, 4, 5\}, \{1, 4, 5, 6\}\}$$

$$M_7 = \{B \mid f^{-1}(B) = \{3, 4, 5, 6\}\} = \{\{2, 3, 4, 5\}, \{2, 3, 4, 5, 6\}\}$$

$$M_8 = \{B \mid f^{-1}(B) = \{1, 2, 3, 4, 5, 6\}\} = \{\{1, 2, 3, 4, 5\}, \{1, 2, 3, 4, 5, 6\}\}$$

- (b) The function f is not measurable with respect to $\mathcal{F}_1 - \mathcal{F}_2$, as, for example, $f^{-1}(\{2, 3\}) = \{1, 2, 3\} \notin \mathcal{F}_1$.
- (c) The image $f(\mathcal{F}_1)$ is not a σ -algebra because $\Omega_2 \notin f(\mathcal{F}_1)$ and also $f(\{3, 4\}) \in f(\mathcal{F}_1)$, but $f(\{2, 3\}) = \{1, 4, 5, 6\} \notin f(\mathcal{F}_1)$.

BLATT 4/2

Given the random variable $X : \Omega \rightarrow (0, \infty)$ on a probability space $(\Omega, \mathcal{F}, P)$ with $X \sim \text{Ga}(1.5, 4)$.

- (a) Is $Y = \sqrt{X}$ a random variable? Justify your answer!

Step-by-Step Analysis

- (1) **Random Variable X :** X is defined as a random variable, taking values from the interval $(0, \infty)$ and following a specific probability distribution, here a Gamma distribution with parameters 1.5 and 4.
- (2) **Transformation via $g(x) = \sqrt{x}$:** The function $g(x) = \sqrt{x}$ is a function that assigns the square root of x to each $x > 0$. This function is continuous and well-defined for all $x \in (0, \infty)$.
- (3) **Measurability of g :** A function g is measurable with respect to the Borel σ -algebra on $(0, \infty)$ if, for every open set $U \subset (0, \infty)$, the preimage $g^{-1}(U)$ is also a Borel set. Since $g(x) = \sqrt{x}$ is continuous, it is also measurable. This implies that the preimages of open sets under g are open (and thus Borel) sets.

- (4) **Construction of $Y = g(X)$:** Since X is a random variable and g is a measurable function, the composition $Y = g(X) = \sqrt{X}$ is also a random variable. This follows from the fact that the measurability of a function is preserved when it is composed with another measurable function.

Why is Y a Random Variable?

The measurability of Y is ensured by the composition of the measurable function g with the random variable X . In measure theory, a function Y is a random variable if it is measurable with respect to the σ -algebra of the probability space, i.e., the preimage of any Borel set under Y must belong to the underlying σ -algebra. Since g maps Borel sets to Borel sets and X is measurable with respect to the σ -algebra, Y is measurable and thus a random variable.

BLATT 4/1

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A, B \in \mathcal{F}$.

- (a) Prove or disprove: A and B are disjoint $\Rightarrow A$ and $\bar{B}$ are stochastically independent.
- (b) Let $P(A) = 0.5$ and $P(A \cup B) = 0.7$. Determine $P(B)$ when:
 - A and B are stochastically independent,
 - A and B are disjoint.
- (c) Consider the event A : "The result is an even number" when rolling a fair die once. Provide an event $B \in \mathcal{P}(\Omega)$ that is independent of A with $B \notin \{\emptyset, \Omega\}$.

Solution

- (a) **Counterexample: Single dice roll**

$$A : \{1, 2\}$$

$$B : \{3, 4\}$$

$$\bar{A} : \{3, 4, 5\}$$

$$\bar{B} : \{1, 2, 5\}$$

$$A \cap \bar{B} : \{1, 2\}$$

$$P(A) = \frac{1}{3}, \quad P(\bar{B}) = \frac{2}{3}, \quad \text{and} \quad P(A \cap \bar{B}) = \frac{1}{3}$$

$\Rightarrow A$ and B are disjoint, but A and $\bar{B}$ are not stochastically independent.

BLATT 4/2

On a measurable space $(\Omega, \mathcal{F})$, let $f : \Omega \rightarrow \mathbb{R}$ be a real function. Demonstrate with an example that the measurability of $|f|$ does not generally imply the measurability of the numerical function f .

BLATT 4/3

Let $X \sim \text{Geo}(p)$, meaning $X : \Omega \rightarrow \mathbb{N}$ is geometrically distributed with $\mathbb{P}(X = k) = p(1-p)^{k-1}$ and $p \in (0, 1)$. Show that

$$\mathbb{P}(\{X = n + k\} \mid \{X > n\}) = \mathbb{P}(X = k)$$

for all $n, k > 0$.

Hint: This property (1) is called the "memorylessness" of the geometric distribution.

BLATT 5/1

For $\lambda > 0$, let the function $F : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - \exp(-\lambda x) & \text{if } x \geq 0. \end{cases}$$

Show that F is a valid cumulative distribution function. *Hint:* Show that F satisfies Definition 5.5. *Note:* The function F defined above is the cumulative distribution function of the exponential distribution (more on this later, in Chapter 10) with parameter $\lambda > 0$. For a random variable X with this distribution function, we also write $X \sim \text{Exp}(\lambda)$.

Solution.

To show that $F(x)$ is a valid cumulative distribution function, we need to verify the following properties:

- (1) $F(x)$ is non-decreasing.
- (2) $F(x)$ is right-continuous.
- (3) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

1. Non-decreasing: We need to show that for any $x_1 < x_2$, $F(x_1) \leq F(x_2)$.

$$\text{If } x_1 < 0 \text{ and } x_2 < 0, F(x_1) = F(x_2) = 0.$$

$$\text{If } x_1 < 0 \text{ and } x_2 \geq 0, F(x_1) = 0 \text{ and } F(x_2) = 1 - \exp(-\lambda x_2) \geq 0.$$

$$\text{If } x_1 \geq 0 \text{ and } x_2 \geq 0, F(x_1) = 1 - \exp(-\lambda x_1) \leq 1 - \exp(-\lambda x_2) = F(x_2).$$

Thus, $F(x)$ is non-decreasing.

2. Right-continuous: We need to check that $F(x)$ is right-continuous, i.e., $\lim_{x \rightarrow x_0^+} F(x) = F(x_0)$ for all $x_0 \in \mathbb{R}$.

$$\text{For } x_0 < 0, F(x_0) = 0 \text{ and } \lim_{x \rightarrow x_0^+} F(x) = 0.$$

$$\text{For } x_0 \geq 0, \lim_{x \rightarrow x_0^+} F(x) = 1 - \exp(-\lambda x_0) = F(x_0).$$

Thus, $F(x)$ is right-continuous.

3. Limit properties:

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1.$$

Since $F(x)$ satisfies all three properties, it is a valid cumulative distribution function.

BLATT 5/2

Let X be the random variable "Net profit (=Profit - Stake)" in a one-time lottery game "6 out of 49." Can you apply the transformation $Y = g(X)$ using Theorem 5.8 in the following situations? If not, provide a brief justification.

- (a) $g(x) = v + x$, where $v \in \mathbb{R}$ is the wealth before the lottery, and thus Y represents the wealth after the lottery.
- (b) $g(x) = \text{sign}(x)$, so Y encodes whether the net result of the game is a win, neither win nor loss, or a loss.
- (c)

$$g(x) = \begin{cases} x & \text{if } x \leq 10 \\ 20 - (x - 10) & \text{if } 10 < x < 20 \\ x & \text{if } x \geq 20. \end{cases}$$

Solution

- (a) **Yes, you can apply Theorem 5.8.** The function $g(x) = v + x$ is a linear transformation, which is one-to-one and continuous. Therefore, Theorem 5.8 applies, and the distribution of Y can be obtained by shifting the distribution of X by v .
- (b) **No, you cannot apply Theorem 5.8.** The function $g(x) = \text{sign}(x)$ is not one-to-one, as it maps multiple values of x to the same value of Y . Therefore, Theorem 5.8 does not apply.
- (c) **No, you cannot apply Theorem 5.8.** The function $g(x)$ defined piecewise is not one-to-one or monotonic over the entire domain of x . Specifically, in the interval $10 < x < 20$, $g(x)$ is decreasing, which violates the conditions required by Theorem 5.8. Hence, Theorem 5.8 does not apply.

BLATT 5/3

Consider the following discrete distributions: Binomial distribution, Geometric distribution, Negative Binomial distribution, and Hypergeometric distribution. Which distribution corresponds to which urn model (considering with/without replacement)? Does the order of drawing play a role in each case?

Solution.

1. Binomial Distribution Urn Model: Sampling with replacement.

Explanation: The Binomial distribution models the number of successes in a fixed number of independent Bernoulli trials, where each trial has the same probability of success. This corresponds to drawing with replacement because the probability of success remains constant for each draw.

2. Geometric Distribution Urn Model: Sampling with replacement.

Explanation: The Geometric distribution models the number of trials needed to get the first success in a series of independent Bernoulli trials. Since each trial is independent and has the same probability of success, this is akin to sampling with replacement.

3. Negative Binomial Distribution Urn Model: Sampling with replacement.

Explanation: The Negative Binomial distribution generalizes the Geometric distribution by modeling the number of trials needed to achieve a fixed number of successes. This is also based on independent trials, so it corresponds to sampling with replacement.

4. Hypergeometric Distribution Urn Model: Sampling without replacement.

Explanation: The Hypergeometric distribution models the number of successes in a fixed number of draws from a finite population without replacement. Here, the probability of success changes with each draw because the population size decreases.

Does the Order Matter?

- **Binomial, Geometric, and Negative Binomial Distributions:** The order does not matter because each trial is independent, and the probability of success remains the same.
- **Hypergeometric Distribution:** The order matters because each draw affects the population, changing the probability of success for subsequent draws.

BLATT 5/4

Which distributions do the following random variables have? Additionally, provide the probability mass function P and the support as precisely as possible.

- (a) In an area lives an unknown number N of animals. To estimate the population size, ecologists use the following procedure: First, they capture a number M of animals and mark them. These are released again. One waits until they have mixed well with the rest and then (randomly) captures n animals. Assuming all animals are caught with equal probability, let X_1 be the number of marked animals in the sample.

- (b) A lump of a radioactive substance consists of many atoms, which emit α -particles upon their very rare decay. Let X_2 be the number of emitted α -particles per time interval.
- (c) A drunk night watchman has a keyring with 10 keys and wants to unlock a door where exactly one of these keys fits. He randomly selects a key. If it doesn't fit, he drops the keyring, the keys get shuffled, and he repeats his approach. Let X_3 be the number of attempts until he finds the correct key.
- (d) Suppose a person from Munich knows every 1000th inhabitant personally. Let X_4 be the number of acquaintances he meets on a walk when he encounters 50 people from Munich.

Solution.

- (a) **Distribution:** Hypergeometric Distribution.

PMF:

$$P(X_1 = k) = \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, \dots, \min(M, n).$$

Support: $k \in \{0, 1, \dots, \min(M, n)\}$.

- (b) **Distribution:** Poisson Distribution.

PMF:

$$P(X_2 = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Support: $k \in \{0, 1, 2, \dots\}$.

- (c) **Distribution:** Geometric Distribution.

PMF:

$$P(X_3 = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

Support: $k \in \{1, 2, 3, \dots\}$.

- (d) **Distribution:** Binomial Distribution.

PMF:

$$P(X_4 = k) = \binom{50}{k} p^k (1 - p)^{50-k}, \quad k = 0, 1, 2, \dots, 50,$$

Support: $k \in \{0, 1, 2, \dots, 50\}$.

BLATT 7/1

Given the random variable $X : \Omega \rightarrow (0, \infty)$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $X \sim \text{Ga}(1.5, 4)$.

- a) Is $Y = \sqrt{X}$ a random variable? Justify your answer.
- b) Compute the expectation of Y .
- Hint:* $\Gamma(1.5) = \frac{\sqrt{\pi}}{2}$.
- c) Show that

$$\text{Var}(Y) = \frac{3}{8} - \frac{1}{\pi}.$$

Part (a) - Is $Y = \sqrt{X}$ a random variable?

Yes, $Y = \sqrt{X}$ is a random variable. This can be justified as follows:

- X is a random variable defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, meaning it is a measurable function from Ω to $(0, \infty)$.
- The function $f(x) = \sqrt{x}$ is continuous and measurable on $(0, \infty)$.
- Since the composition of a measurable function f with a random variable X is also a random variable, $Y = f(X) = \sqrt{X}$ is indeed a random variable.

Part (b) - Compute the expectation of Y

We are asked to compute $\mathbb{E}[Y] = \mathbb{E}[\sqrt{X}]$, where $X \sim \text{Ga}(1.5, 4)$. The probability density function (PDF) of a Gamma random variable $X \sim \text{Ga}(\alpha, \beta)$ is given by:

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

For $X \sim \text{Ga}(1.5, 4)$, this becomes:

$$f_X(x) = \frac{4^{1.5}}{\Gamma(1.5)} x^{0.5} e^{-4x}$$

Now, the expectation of $Y = \sqrt{X}$ is:

$$\mathbb{E}[Y] = \int_0^\infty \sqrt{x} f_X(x) dx$$

Substituting the PDF:

$$\mathbb{E}[Y] = \int_0^\infty \sqrt{x} \cdot \frac{4^{1.5}}{\Gamma(1.5)} x^{0.5} e^{-4x} dx$$

Simplifying:

$$\mathbb{E}[Y] = \frac{4^{1.5}}{\Gamma(1.5)} \int_0^\infty x e^{-4x} dx$$

Using the known result $\int_0^\infty x e^{-\lambda x} dx = \frac{1}{\lambda^2}$ for $\lambda = 4$:

$$\mathbb{E}[Y] = \frac{4^{1.5}}{\Gamma(1.5)} \cdot \frac{1}{4^2}$$

Now, simplify the expression:

$$\mathbb{E}[Y] = \frac{4^{1.5}}{\Gamma(1.5)} \cdot \frac{1}{16} = \frac{8}{\frac{\sqrt{\pi}}{2}} \cdot \frac{1}{16} = \frac{1}{\sqrt{\pi}}$$

Thus, the expectation of Y is:

$$\mathbb{E}[Y] = \frac{1}{\sqrt{\pi}}$$

Part (c) - Variance of Y

To compute the variance of $Y = \sqrt{X}$, we need $\mathbb{E}[Y^2]$ and $\mathbb{E}[Y]^2$.

Since $Y = \sqrt{X}$, we have $Y^2 = X$, and the expectation of Y^2 is simply the expectation of X , which is known for a Gamma distribution:

$$\mathbb{E}[Y^2] = \mathbb{E}[X] = \frac{\alpha}{\beta} = \frac{1.5}{4} = \frac{3}{8}$$

Now, we already know from part (b) that:

$$\mathbb{E}[Y] = \frac{1}{\sqrt{\pi}}$$

Thus, $\mathbb{E}[Y]^2 = \left(\frac{1}{\sqrt{\pi}}\right)^2 = \frac{1}{\pi}$.

Finally, we can calculate the variance of Y :

$$\text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 = \frac{3}{8} - \frac{1}{\pi}$$

Therefore, the variance of Y is:

$$\text{Var}(Y) = \frac{3}{8} - \frac{1}{\pi}$$

BLATT 7/2

Let X be a random variable over $(\Omega, \mathcal{F}, \mathbb{P})$ with distribution P_X , and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$g(x) = x \cdot \mathbb{I}_{[-2,2]}(x) + 2 \cdot \mathbb{I}_{(2,\infty)}(x) - 2 \cdot \mathbb{I}_{(-\infty,-2)}(x).$$

- (a) Justify why $g(X)$ is a random variable. As always, a sketch may be helpful.
 - (b) Show (e.g., with an appropriate estimation) that the expected value of the random variable $g(X)$ exists, i.e., $\mathbb{E}(g(X)) < \infty$.
 - (c) Show that $\mathbb{E}(g(X)) = 0$ if P_X has a density f with respect to the Lebesgue measure λ (i.e., $P_X = f \odot \lambda$), where f is symmetric around 0.
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